RAPE CRIME STATISTICS: A COMPARATIVE STUDY AMONG THREE NATIONS

Report submitted to

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For the award of the degree

of

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by

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This is to certify that **MANJIMA RIT** (**Registration No:** 220010044170) has prepared the project work entitled **RAPE CRIME STATISTICS: A COMPARATIVE STUDY AMONG THREE NATIONS** under the supervision of **Prof. (Dr.) ANINDITA GHOSAL** based on the survey of literatures in his area of interest for the partial fulfilment of the B.Sc. (Hons.) Degree in Statistics from Sister Nivedita University.

Ghosal

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Assistant Professor and HOD

Arindam Gupta 2/5/25

Signature of External with date

Ghosal 2/5/2025

Signature of Supervisor with date

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DECLARATION BY THE STUDENT

I hereby declare that the work which is being presented in this dissertation entitled "**RAPE CRIME STATISTICS: A COMPARATIVE STUDY AMONG THREE NATIONS**" in partial fulfilment of the requirements for the award of the degree of Bachelors/Masters/Ph.D. of **Science in Statistics** is an authentic record of my own work carried out during the period from **6th January 2025** to **27th April 2025** under the supervision of **Prof. (Dr.) Anindita Ghosal, Department of Statistics, Sister Nivedita University**.

I acknowledge that I have read and understood the UGC regulations, 2018 (*Promotion of Academic integrity and Prevention of Plagiarism in Higher Educational Institutions*).

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Manjima Rit 2/5/25

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Contents

Serial No.		Topic	Page No.
I.		Abstract	6
II.		Introduction	7
III.		Objective	8
IV.		Methodology	9
V.		Datasets	10-12
VI.		Analysis	
	i.	Graphical Representation	13-18
	ii.	Quality of the Data	19-23
	iii.	Global Analysis with Model Fitting and Forecasting	24-28
	iv.	Comparison of the three countries	29
	v.	Analysis and Model Fitting of the of 10 states in India	30-50
	vi.	Future ranking the states based on the forecasted values	51
VII.		Conclusion	52-53
VIII.		Limitations & Future works	54
IX.		Appendix	55-60
X.		Reference	61-62
XI.		Data source	63

Abstract

This study undertakes a comparative, on statistical ground for one of the most pressing and sensitive social issues faced world-wide that is rape. Focusing on three diverse countries, namely India, Germany and Kenya, the study aims to analyse patterns, trends, and societal impacts by utilizing official crime statistics data drawn from national and international crime reports. Using time series analysis and comparative data visualization techniques, the project evaluates the reported rape cases across these countries over the past two decades, measuring rates per 100,000 population and examining year-on-year growth or decline.

Within India, the study delves deeper by exploring rape crime rates across various states and union territories. Based on an average analysis of the past 19 year's data, we choose 10 states (i.e.; Delhi, Mizoram, Madhya Pradesh, Assam, Arunachal Pradesh, Chhattisgarh, Rajasthan, Tripura, Chandigarh and Haryana) that consistently reflect high case volumes due to rape crime. It further provides predictive insights into the future trends of reported rape cases for the next 10 years (2023 to 2032) for these Indian states and also for Germany, Kenya and India based on their total rape crime rate.

Beyond mere statistical analysis, this research also seeks to shed light on the underlying systemic and structural issues contributing to the persistent prevalence of rape. By combining statistical forecasting with socio-cultural critique, the study emphasizes the urgent need to address gender-based violence on a global scale, providing critical insights into how this crime is evolving over time and the societal transformations and strict laws required to combat it.

Introduction

General Legal definition of Rape: Rape is the act of engaging in sexual intercourse or other forms of sexual penetration with another person without their consent, typically through force, threats, manipulation or when the person is incapable of giving consent due to age, mental incapacity, or intoxication.

Rape is one of the most heinous and traumatic crimes that deeply scars not only the victim but also the collective conscience of society. This research is to draw attention towards the globally increased Rape Crime in the world. By a comparative study among three countries of the world we aim to find if this crime is regional, economical, educational or societal based. Firstly, Germany (a European country) which is considered to be a developed country in economic field. Secondly, Kenya (an African country) which is considered to be a lower- or middle-income country in economic field. Thirdly, India (an Asian country) which is considered to be a developing or lower- or middle-income country in economic field and most importantly it's our nationality. While there has been growing discourse around women's safety in India, there remains a gap in long-term predictive research that can aid proactive policy and societal response.

Moreover, considering mainly India, this project further explores with the Top 10 States and Union Territories dealing with their Rape Crime Rates, comparing the states with each other. Divining into the brief details of India, we see despite the existence of strict laws and increasing awareness around women's rights and safety, rape remains a grave and persistent issue. The country has and continues witness a troubling rise in reported rape cases over the years, highlighting deep-rooted issues such as gender inequality, social stigma, lack of education and gaps in law enforcement. The issue gained global attention after the 2012 Nirbhaya gang rape case in Delhi, which led to widespread protests and demands for reform. In response, the Indian government introduced several legal changes, including stricter punishments and the establishment of fast-track courts. However, the prevalence of such crimes continues and many cases still go unreported due to fear of social ostracism, victim-blaming, and lack of faith in the justice system.

Objective

The primary objective of this research is to address one of the most alarming issues faced world-wide by women today that is the prevalence of rape. Analyse and compare the rape crime data of 3 countries (India, Germany & Kenya) across the world. It mainly compares the various (Top 10) states in India using forecasting models such as time series plots, data visuals and ARIMA (Auto-Regressive Integrated Moving Average) by doing so, the study seeks to:

I. **To Analyse Historical Trends of Rape Cases**: The study examines the year-wise data on reported rape cases of the TOP 10 Indian states and union territories. Also, examines the global trend year-wise of total cases for 3 countries. Identifies patterns, fluctuations, and significant spikes or drops in reported cases for these countries.

II. **Identify State-wise Hotspots of Gender-based Violence**: Helps to highlight the regions with consistently high or increasing numbers of rape cases and compares the state performances based on normalized crime rates (cases per lakh population) which helps to understand socio-demographic factors contributing to regional disparities.

III. **To Forecast Rape Cases for the Next 10 Years (2023–2032)**: We have used time series models like ARIMA to forecast the number of rape cases in each of the (TOP 10) states. The prediction of long-term trends helps to assess whether the situation is likely to improve or worsen without significant intervention.

IV. **Data Visualisation causes public awareness:** Through interactive visualizations and accessible analysis, the project can educate the public about the ongoing and projected state of women's safety in India. It can help shift the conversation from reactionary outrage to sustained, preventive action.

Methodology

The methods used while doing the research are as follows:

I. **Data Analysis through Graphical Representations:** Here we are going to represent the dataset graphically viz. dataset corresponding to the state-wise rape crime in **India**, **Germany** and **Kenya** depends upon Rape Crime Rate cases and different factors (like age of the victims and relation with the offender) for the years **2004-2022**. The datasets corresponding to the past 19 years of rape crime rate cases data for India and the datasets corresponding to the past 19 years of rape crimes data for India through the factors. And we are going to represent all this data set with some statistical tools like **line charts** and **heatmaps.**

II. **Quality checking of the data:** To proceed with the given datasets first we have to check whether the datasets are fit for our analysis. So, with the help of **boxplots** we check for the **outliers** as it is crucial in data analysis because outliers can significantly distort statistical results and lead to incorrect conclusions.

III. **Time Series Forecasting:** Here we are going to forecast our dataset corresponding to the past 19 years of Rape Crime rate data for Germany, Kenya, India and its states and the datasets corresponding to the past years 2004-2022 and try to get the forecasted values for the upcoming next 10 years (i.e., 2023-2032). For predicting the future values, we have used forecasting models like ARIMA (Auto-Regressive Integrated Moving Average) or to be specified auto. Arima () test in R-programming which is known for fitting the best ARIMA model to the given data with lowest AIC (Akaike Information Criterion) and deals well with the data having non-seasonality and also converts the non-stationary data to stationary to work with it.

IV. **Comparison of the three countries:** In this step, we are going to compare the forecasted values for upcoming 10 years which we can get from previous step and compare on basis of that.

V. **Prediction of the rank of Top 10 states in India:** Here we are going to forecast the dataset corresponding to the past 19 years Rape Crime Rate data across all the states for India and then select only the top 10 states corresponding to the past 19 years data and forecast them. After we get the forecasted values, we can rank them as TOP 10 state corresponding to forecasted values for upcoming 10 years (2023-2032).

Dataset

I. Dataset of **India** based on Rape Crime Rate for **2004-2022** years.

Table-1

Years	Rape Crime Rate of India
2004	1.7
2005	1.7
2006	1.7
2007	1.8
2008	1.9
2009	1.8
2010	1.9
2011	2
2012	4.26
2013	5.69
2014	6.1
2015	5.7
2016	6.3
2017	5.2
2018	5.2
2019	4.3
2020	4.3
2021	4.8
2022	4.7

II. Dataset of **Germany** based on Rape Crime Rate for **2004-2022** years.

Table-2

Years	Rape Crime Rate of Germany
2004	10.87
2005	10.01
2006	10
2007	9.25
2008	8.98
2009	9
2010	9.5
2011	9.26
2012	9.85
2013	9.07
2014	8.97
2015	8.56
2016	9.62
2017	10.06

2018	10.9
2019	11.52
2020	12.15
2021	12.48
2022	15.06

III. Dataset of **Kenya** based on Rape Crime Rate for **2004-2022** years.

Table-3

Years	Rape Crime Rate of Kenya
2004	4.14
2005	3.81
2006	3.5
2007	2.3
2008	1.88
2009	2.1
2010	2.22
2011	2.19
2012	1.8
2013	2.13
2014	1.95
2015	1.82
2016	1.86
2017	1.67
2018	1.96
2019	2.06
2020	1.81
2021	1.78
2022	1.52

IV. Dataset of **(TOP 10) Indian States and Union Territories** based on Rape Crime Rate of **19 years**, considered on total average of each state and union territories.

Table-5

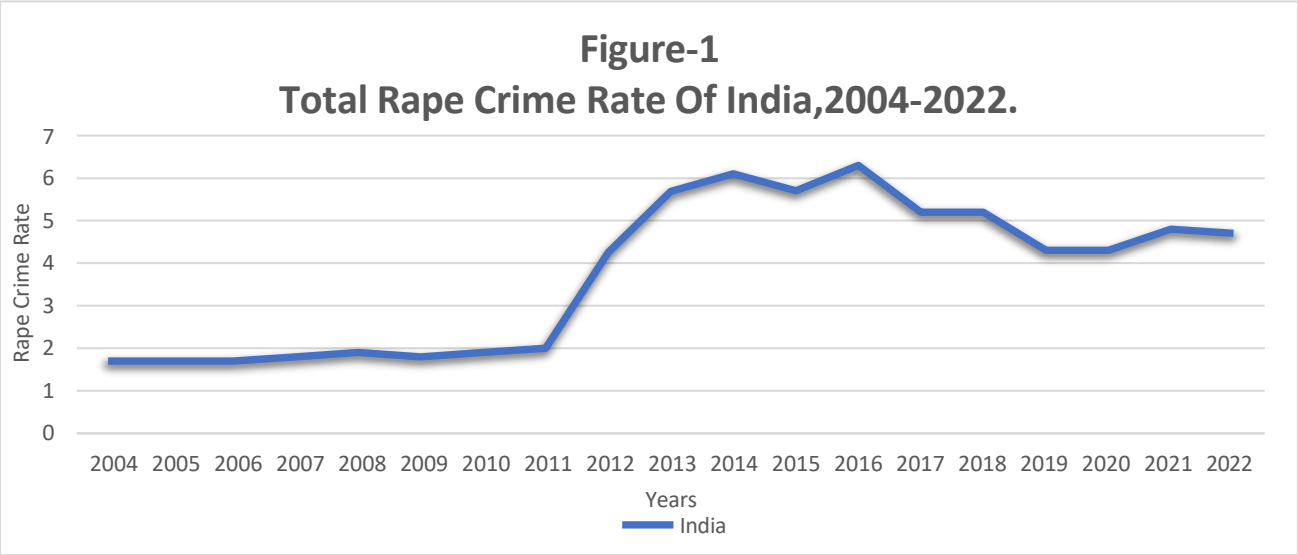
	Delhi	Mizoram	Madhya Pradesh	Assam	Arunachal Pradesh	Chhattisgarh	Rajasthan	Tripura	Chandigarh	Haryana
2004	3.6	2.1	4.4	4.2	3.7	4.4	1.7	4.8	1.9	1.7
2005	4.2	3.9	4.4	4.3	3	4.4	1.6	4.8	3.3	2
2006	3.8	7.5	4.3	4.3	3.2	4.3	1.7	5.5	1.9	2.6
2007	3.6	8.5	4.4	4.9	4	4.2	1.9	4.5	2.1	2.1
2008	2.7	7.8	4.2	4.8	3.5	4.1	2.1	5.8	1.9	2.6
2009	2.6	8.3	4.2	5.3	4.8	4	2.3	5.3	2.6	2.5
2010	2.8	9.1	4.3	5.6	3.8	4.1	2.3	6.6	2.8	2.9
2011	3.4	7.1	4.7	5.5	3	4.1	2.6	5.6	2.6	2.9
2012	8.26	20.81	9.72	11.34	7.71	8.41	6.19	12.77	4.21	5.6

2013	18.63	17.8	12.11	12.65	12.42	11.08	9.78	12.84	6.74	8.02
2014	23.2	23.7	14	12.8	13.6	11.4	11	13	8.5	9.6
2015	23.7	11.3	11.9	11	11.5	12.2	10.5	11.5	10.1	8.6
2016	22.6	4.4	13.1	11.2	14.7	12.6	10.4	11	9.3	9.4
2017	12.5	4.8	14.7	11	9.4	14.6	9.3	5	8.6	8.6
2018	13.3	8.5	13.8	9.9	9.3	14.7	11.7	5	16.1	9.7
2019	10.5	5.5	5.8	9.7	8.1	8.3	13.9	4	10.9	10
2020	10.5	5.5	5.8	9.7	8.1	8.3	13.9	4	10.9	10
2021	12.9	4.3	7.2	10	11.1	7.4	16.4	3	13.3	12.3
2022	12.3	2.3	7.3	6.4	9.8	8.3	13.8	3.1	13.9	12.7

Data Analysis with Graphical Representation

INDIA

I. Line diagram for the dataset of India showing total Rape Crime Rate with of India from 1997-2022 years, which includes all the states and union territories.



From figure- 1, we can observe that:

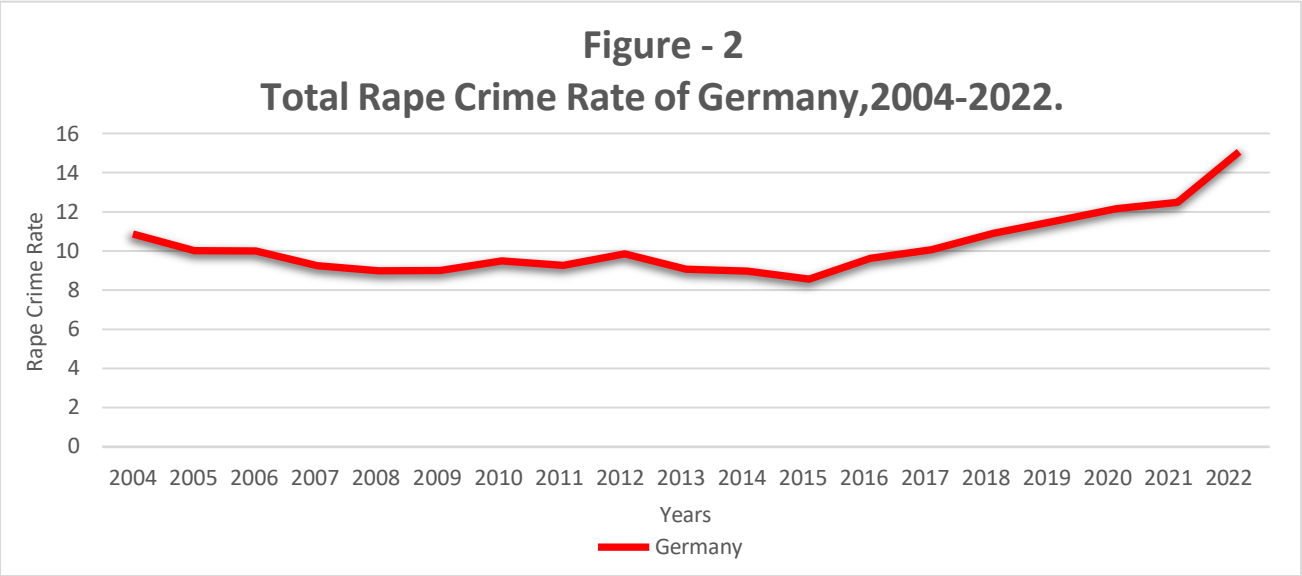
a) India has a low Rape Crime case in the beginning of last 26 years of my dataset but eventually saw its rapid increase from 2004 to 2008 and decreased a little after 2008-2009 but then saw an increase from 2009-2010 and decreased in the year 2011.

b) Cases seems to have increase after 2011 with its peak years as 2012,2013,2014 and again decreased for the year 2015 but then reached the peak of increase in 2016 among all the years.

c) Cases saw a decreased in the years 2017,2019-2020 and slightly decreased in 2021's end although cases remained increased in the years 2018 and 2021.

GERMANY

II. Line diagram for the dataset of Germany showing total Rape Crime Rate per 100,000 per population of Germany from 2004-2022 years.



From figure-2, we can observe that:

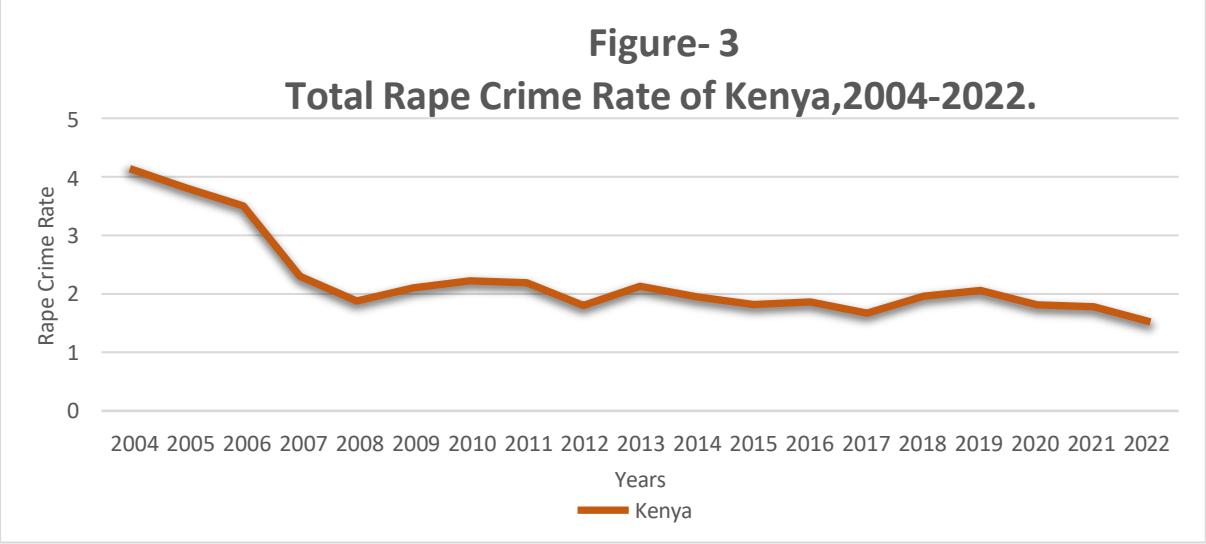
a). Rape crime cases in Germany seems to have an increased graph during the past 20 years starting from 2003 in my dataset.

b). It once started to decrease in the year 2006 to 2009 and gradually again started to increase from 2010 to 2012.

c). Then saw its last time decrease in the years from 2013 to 2015 and took its hike of increasing in the year 2022.

KENYA

III. Line diagram for the dataset of Kenya showing total Rape Crime Rate per 100,000 per population of Kenya from 2004-2022 years.



From figure- 3, we can observe that:

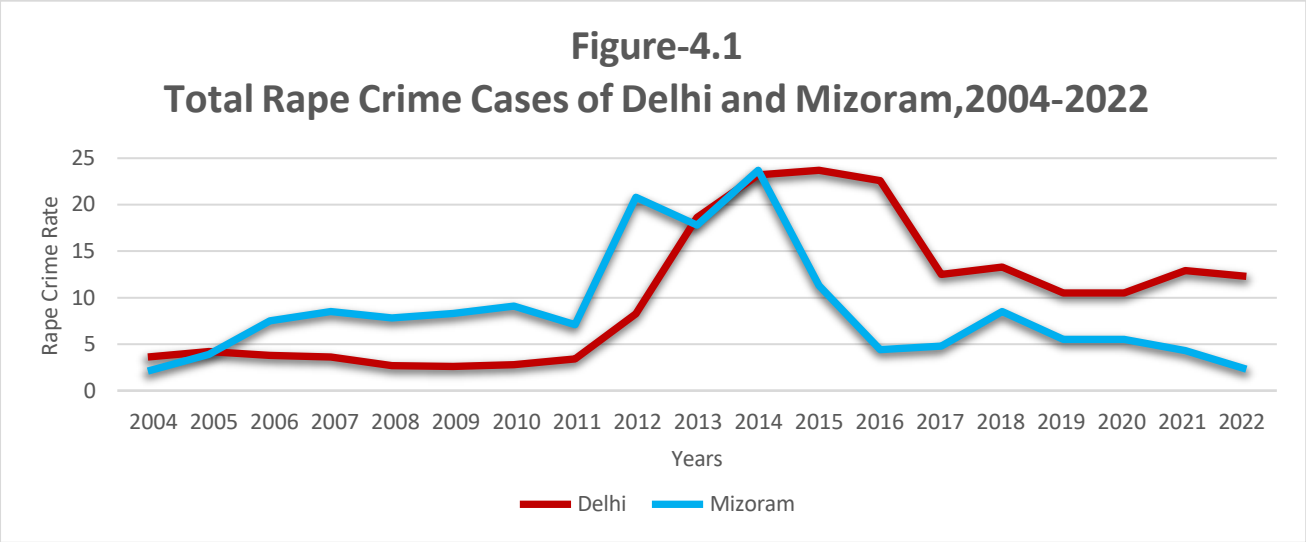
a). Rape crime cases had seen a hike in Kenya in the year 2004.

b). After that, gradually it started to decrease till 2008 but again saw a rise from 2009 to 2011.

c). Again, decreased in the years 2012,2014,2015,2017,2020 and had its minimum reported cases during the years 2021-2022, while increased in the years 2013,2016,2018,2019.

TOP 10 States of India

IV. Line diagram showing the dataset of **10 Indian states and union territories** of past **19 years** having the highest Rape Crime Rate **per lakhs population** ranked on an average in descending order all across India.

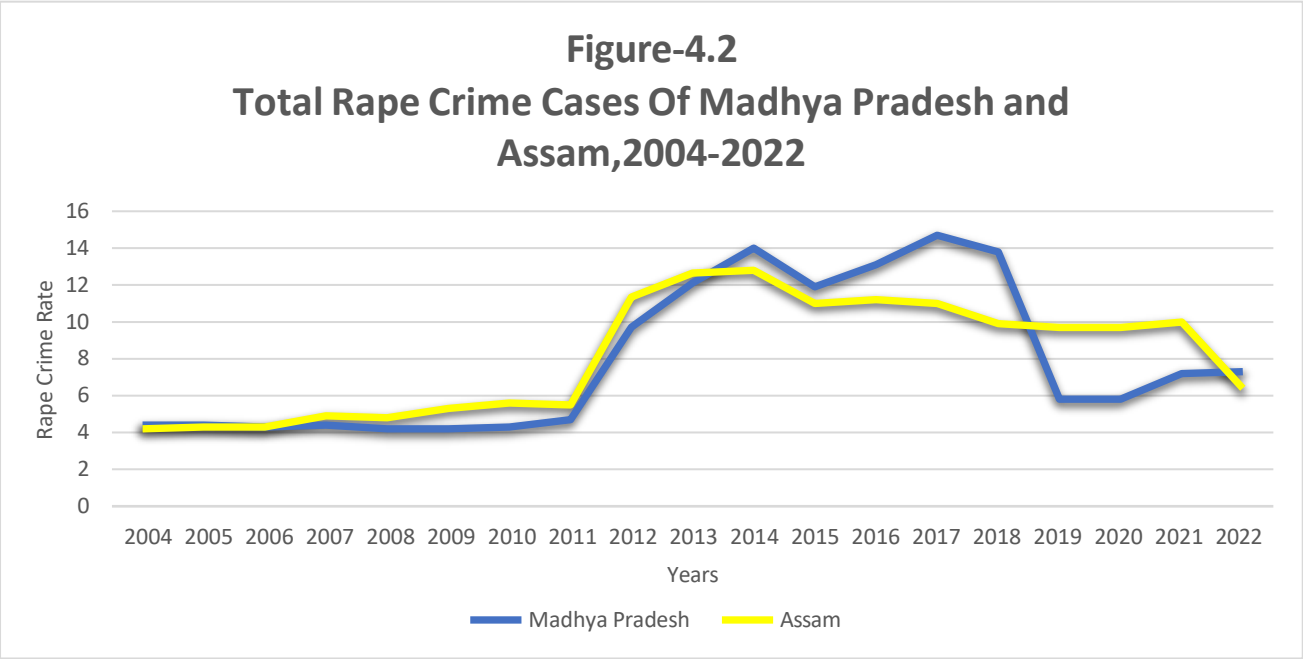


From figure-4.1, we can see:

a) In spite of being so small area-wise Delhi has the highest reported rape cases across all the states and union territories in India followed by Mizoram in the years (2004-2022).

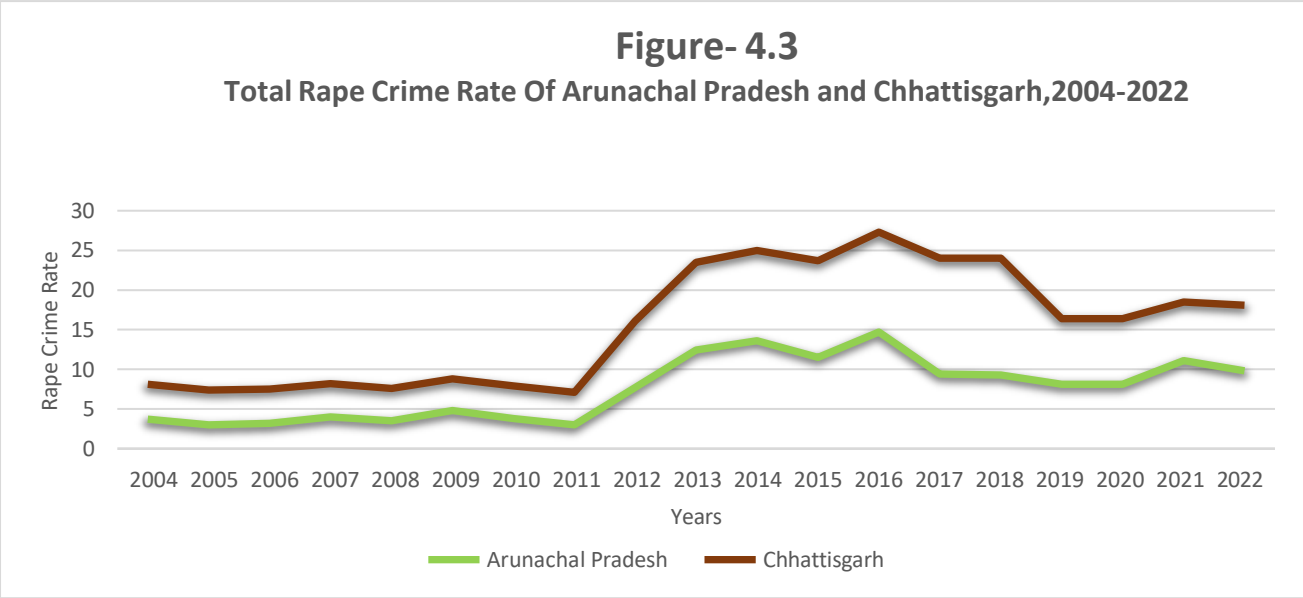
b) From the line diagram it is visible that Delhi in the beginning had lower number of cases compared to Mizoram but cases increased since 2011 and remained increased with short-term fluctuations after 2016.

c) seems to have high cases since 2004 with its highest reported cases in the year 2014 and drastically decreased in the later years.



From figure- 4.2, we can see:

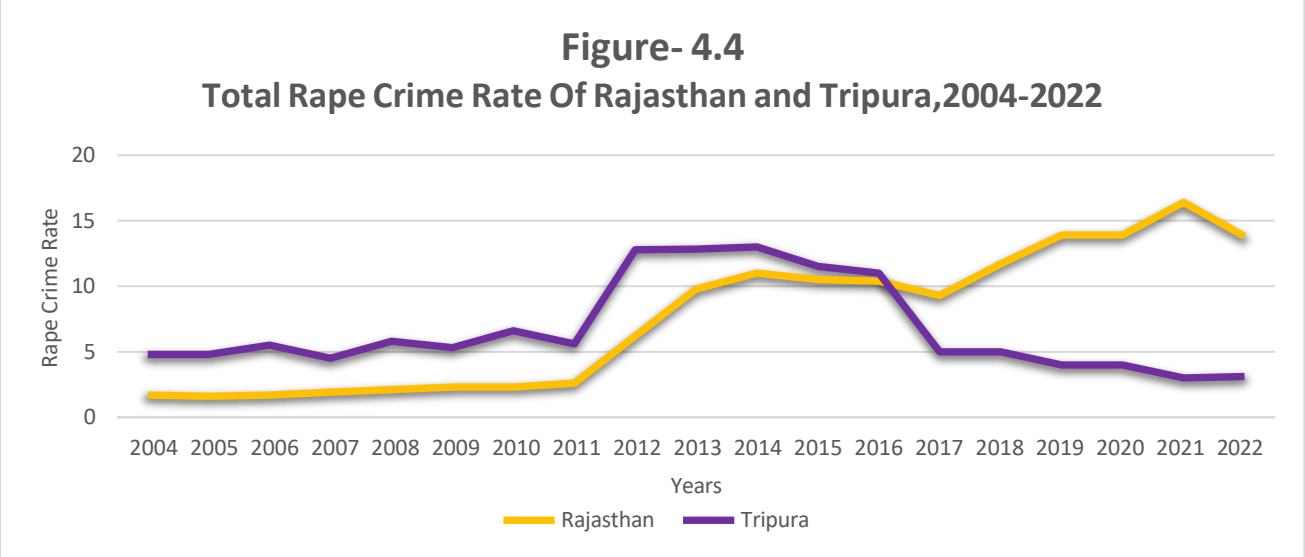
a) Third state with highest reported rape cases in India is Madhya Pradesh and the Assam.
b) Both of the states had lesser reported cases in the beginning but increased in 2012. Assam saw its peak in reported cases in the year 2013 and Madhya Pradesh in 2017.



From figure-4.3, we can see:

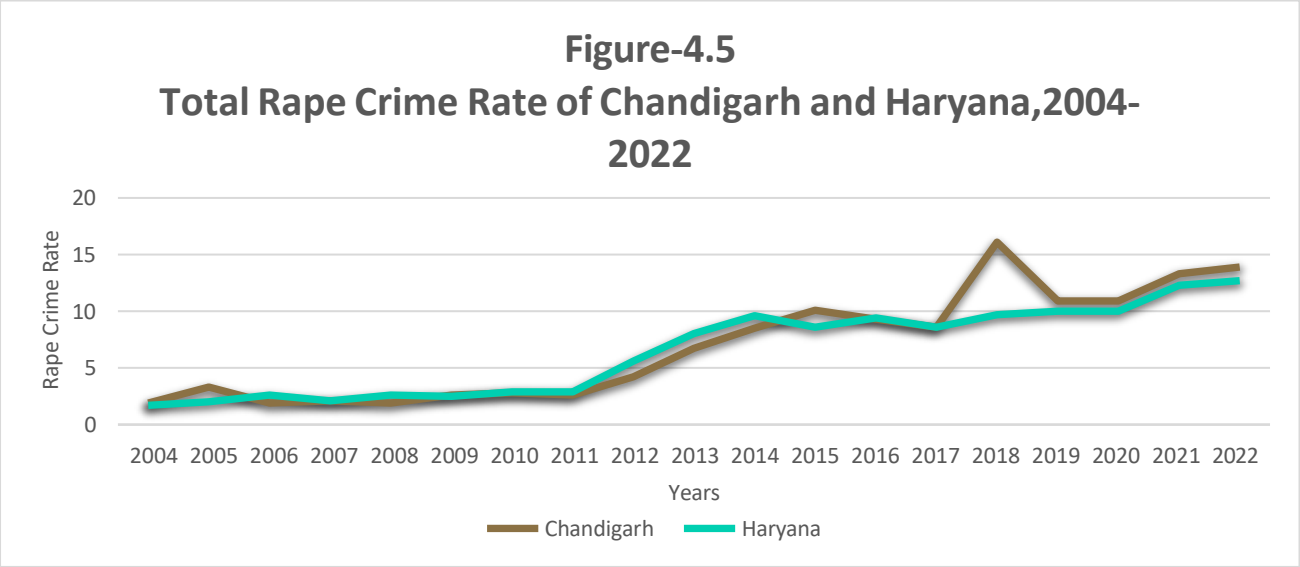
a) Arunachal Pradesh is the fifth top state which has short-term fluctuations visible through the line diagram and followed by Chhattisgarh who had moderately stable growth since the beginning.

b) Arunachal Pradesh saw its highest reported rape cases in the year 2016 and Chhattisgarh saw its highest in 2018. In the later years even though Chhattisgarh decreased (2019-2022), Arunachal Pradesh stabilised just for the years 2019-2020 and rised in the year 2021 despite lockdown conditions across the India.



From figure-4.4, we can see:

a) Rajasthan being the top seventh state with highest reported rape crime followed by Tripura. Rajasthan seems to have a very low reported rape cases from 2004-2011, after which it has an upward line diagram till 2021 with highest reported cases.
b) Tripura has short-term fluctuations since the beginning with highest rise in reported rape cases in the year 2012. Gradually, decreased after 2016 which gives a positive look over.



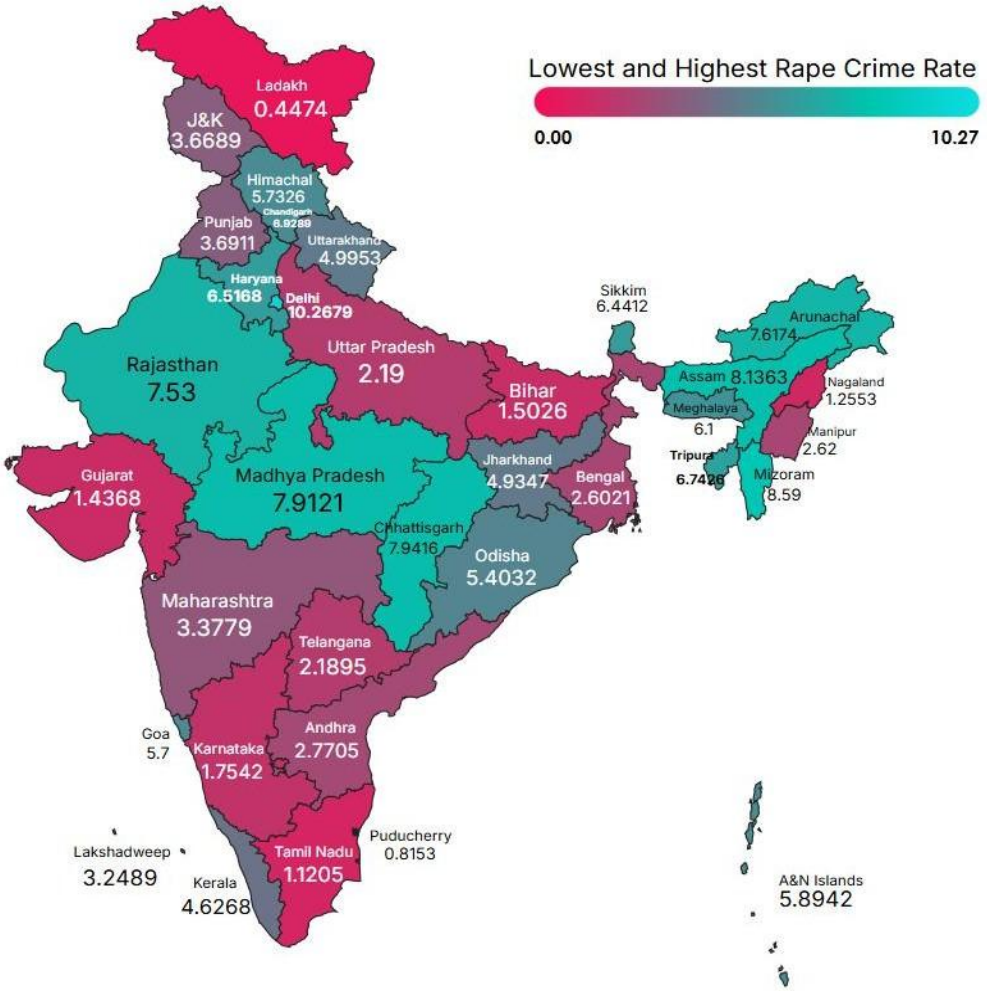
From figure- 4.5, we can see:

a) Chandigarh is the ninth state in the top 10 states of India having highest rape crime rate. Lastly, followed by Haryana.
b) Chandigarh has short-term fluctuations in the beginning of our given years and saw its increase in reported cases in the year 2015 and highest in the year 2018.
c) Haryana too has short-term fluctuations in the beginning of our given years and saw its increase in the year 2014 with its highest reported cases in the year 2021.
d) Even though lockdown prevailed all over the country due to COVID-19 pandemic, in these two states cases continued to increase.

V. **Heatmap** for Rape Crime Rate all over India showing the states and union territories with lowest and highest rape crime rate for the years 2004 to 2022 on an average.

Figure- 5

Heatmap for India showing the states and union territories with highest and lowest rape crime rate,2004-2022.



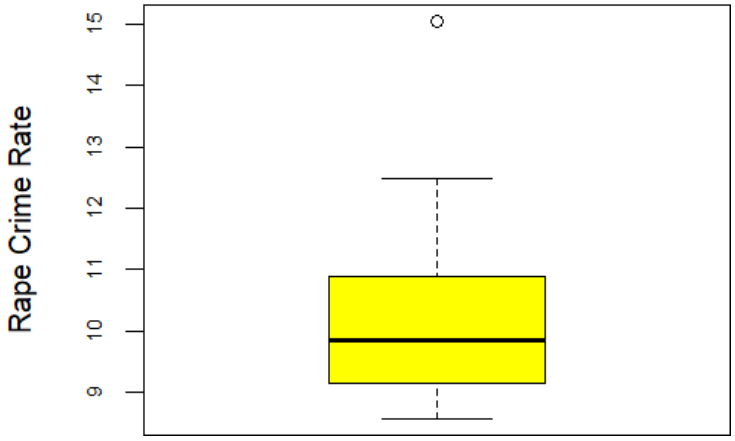
The Quality of the Data

I. Usage of **boxplot** for **outliers** checking in the data.

Outlier - An outlier is a data point that deviates significantly from other data points in a dataset, often appearing as an extreme value or a value that is far from the main cluster of data. Outliers can be exceptionally high or low compared to the other values in the dataset. They can be caused by various factors, including errors in data entry, genuine extreme values, or a mix of both.

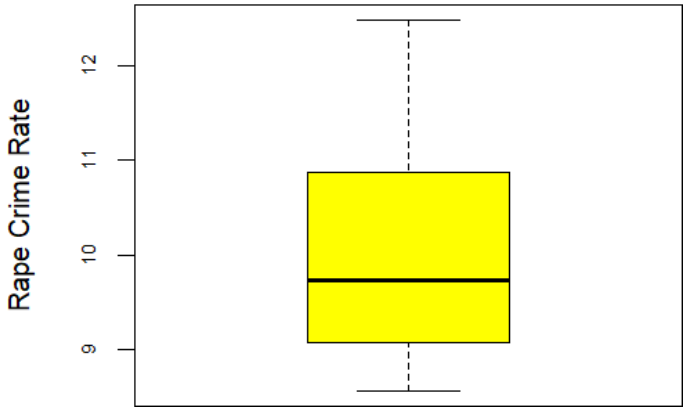
a. Germany

Figure-6
BOXPLOT FOR GERMANY



In figure-6, we can see that the boxplot for Germany has been plotted for the data of Rape Crime Rate for the years 2004 to 2022. As, one data point is outside of the whiskers hence it is considered as outliers and to proceed with our forecasting we remove them.

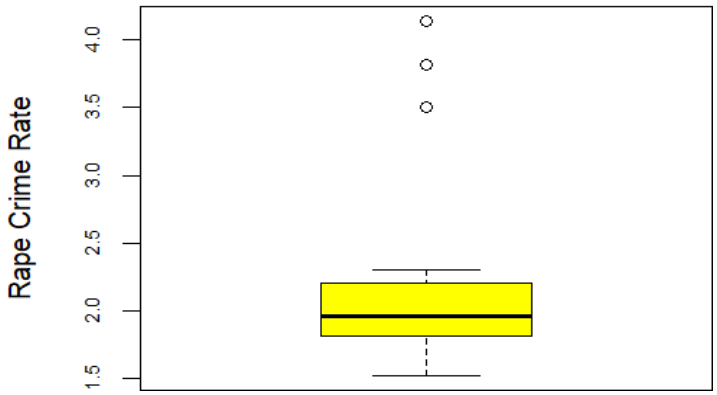
Figure-6.1
BOXPLOT FOR GERMANY



In figure- 6.1, firstly we had 19 data points and one outlier was detected, after first attempt of removal of outliers we see that the no outliers are left so we proceed with this data of ours with 5% of data loss.

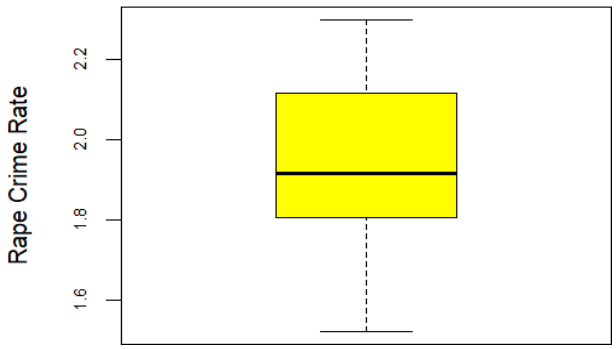
b. Kenya

Figure-7
BOXPLOT FOR KENYA



In figure- 7, we can see that the boxplot for Kenya has been plotted for the data of Rape Crime Rate for the years 2004 to 2022. As, three data points are outside of the whiskers hence it is considered as outliers and to proceed with our forecasting we remove them.

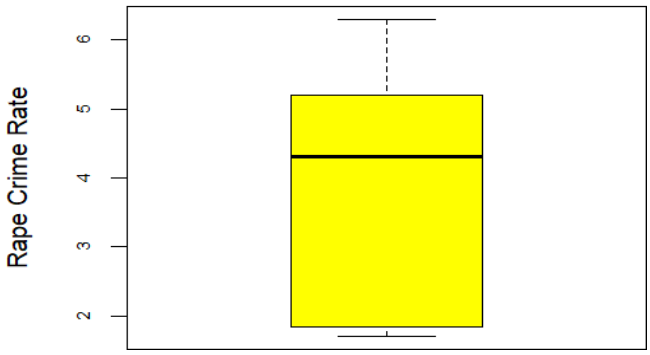
Figure-7.1
BOXPLOT FOR KENYA



In figure- 7.1, firstly we had 19 data points and three outliers were detected, after first attempt of removal of outliers we see that the no outliers are left so we proceed with this data of ours with 15% of data loss.

c. India

Figure-8
BOXPLOT FOR INDIA



In figure- 8, we can see that the boxplot for India has been plotted for the data of Rape Crime Rate for the years 2004 to 2022. As, no data points are outside of the whiskers hence we proceed with the dataset for performing forecasting models.

d. Top 10 Indian states

Figure-9
BOXPLOT FOR DELHI

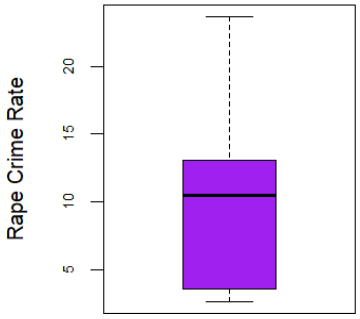


Figure-10
BOXPLOT FOR MIZORAM

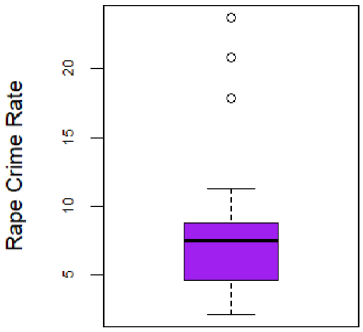


Figure-11
BOXPLOT FOR MADHYA PRADESH

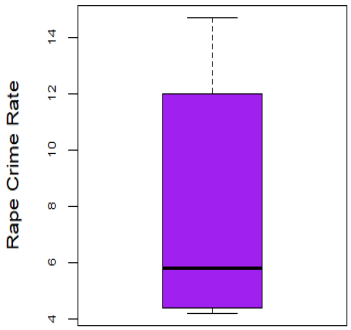


Figure-12
BOXPLOT FOR ASSAM

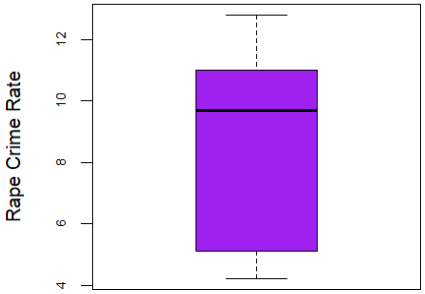


Figure-13
BOXPLOT FOR ARUNACHAL PRADESH

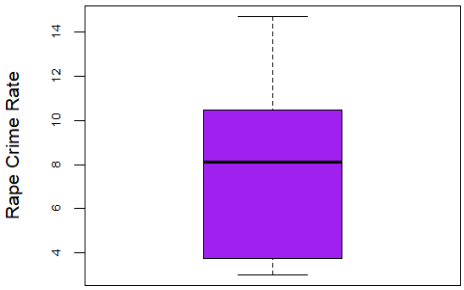


Figure-14
BOXPLOT FOR CHHATTISGARH

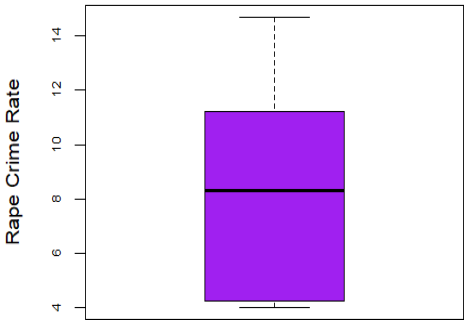


Figure-15
BOXPLOT FOR RAJASTHAN

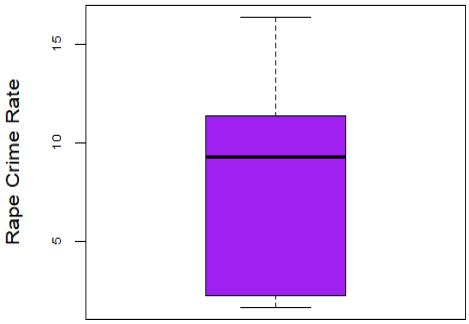


Figure-16
BOXPLOT FOR TRIPURA

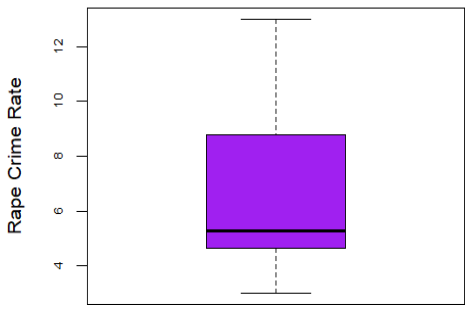


Figure-17
BOXPLOT FOR CHANDIGARH

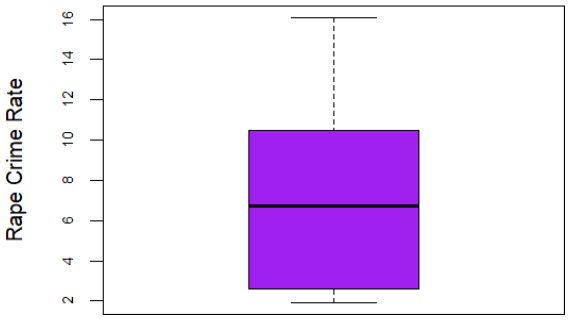
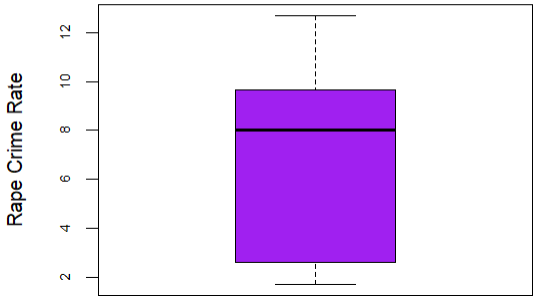
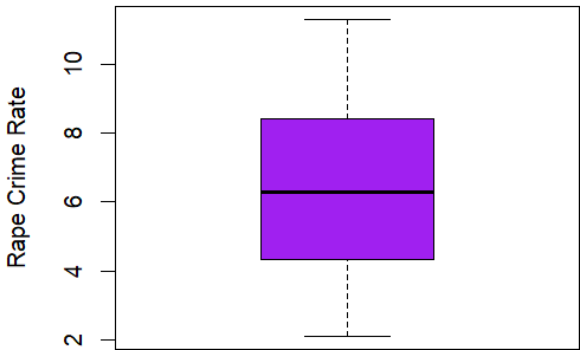


Figure-18
BOXPLOT FOR HARYANA



From all the figures (9-18) we can conclude that the boxplots have been plotted for the data of Rape Crime Rate from 2004-2022 years, of the top 10 states of India and none of the boxplots have outliers except for Mizoram as three data points are outside of the whiskers. Hence, we remove the outliers from the dataset and proceed with performing our forecasting analysis. For the rest of the states as no outliers are present so we proceed with performing our forecasting analysis.

Figure-10.1
BOXPLOT FOR MIZORAM



In figure- 10.1, firstly we had 19 data points and three outliers were detected, after first attempt of removal of outliers we see that the no outliers are left so we proceed with this data of ours with 15% of data loss.

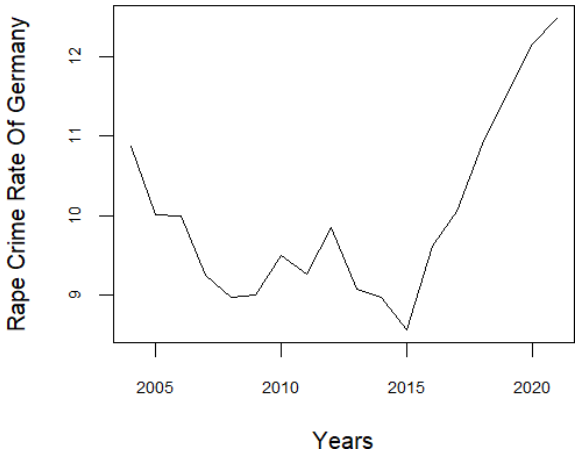
II. Seasonality in the data: By definition of seasonality we know that the period of seasonality is less than a year.
As we have yearly data, seasonality will not exist in this dataset.

Global Analysis with Model Fitting and Forecasting

I. **Forecasting rape crime cases for Germany:** We will try to forecast the data of future values of rape cases based on all over Germany using the data of past 19 years. For forecasting future time points, we have considered the dataset corresponding to the rape crime cases for previous 19 years (2004-2022) for Germany and we are trying to predict the future values for the next 10 years (2023-2032).

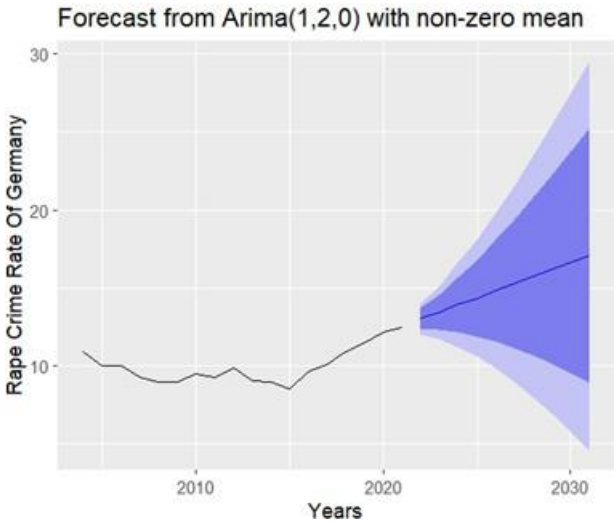
Now, for the said purpose we forecast the future time points by directly using the function auto. Arima () which gives the best fitted model with MA (Moving Average), AR (Auto-Regressive) and differencing parameters, according to lowest AICc (Akaike Information Criterion Corrected) of the given data set.

Figure-19 (Time Series Plot)



From this above time series plot we can conclude that the overall trend is non-linear and exhibits fluctuations, with a noticeable half U-shaped pattern in the first half, followed by a sharp upward trend in the year 2022.

Figure-20 (Forecasting Graph)



From the above forecasting plot, we can conclude that the data fits right in Arima (1,2,0) which is notated as;

$y_t - y_{t-1} = \phi_1(y_{t-1} - y_{t-2}) + \epsilon_t$, where y_t is the time series value at time t, $\boldsymbol{\phi}_1$ is the AR(1) (auto-regressive) coefficient, y_{t-1} and y_{t-2} are the past values of the time series and $\epsilon_t \sim N(0, \sigma 2)$ is white noise error term.

This model forecasts the current value of the differenced series based on its past values (first and second lags) and a random error term.

Forecasted values:

Table: 19.1 showing the forecasted values for the next **10 year (2023-2022)** of **Germany**.

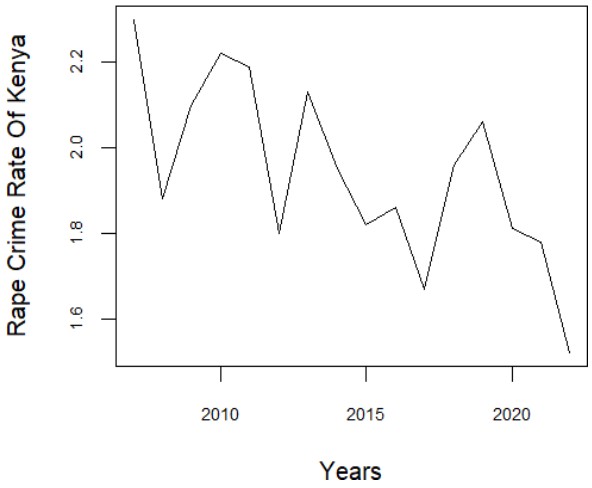
Years	Forecast Points
2023	13.41399
2024	13.91169
2025	14.33503
2026	14.81084
2027	15.24963
2028	15.71454
2029	16.16102
2030	16.6205
2031	17.07081
2032	17.5276

Rape crime rate of Germany is forecasted to moderately increases in the coming years (2023–2032).

II. **Forecasting rape crime cases for Kenya:** We will try to forecast the data of future values of rape cases based on all over Kenya using the data of past 19 years. For forecasting future time points, we have considered the dataset corresponding to the rape crime cases for previous 19 years (2004-2022) for Kenya and we are trying to predict the future values for the next 10 years (2023-2032).

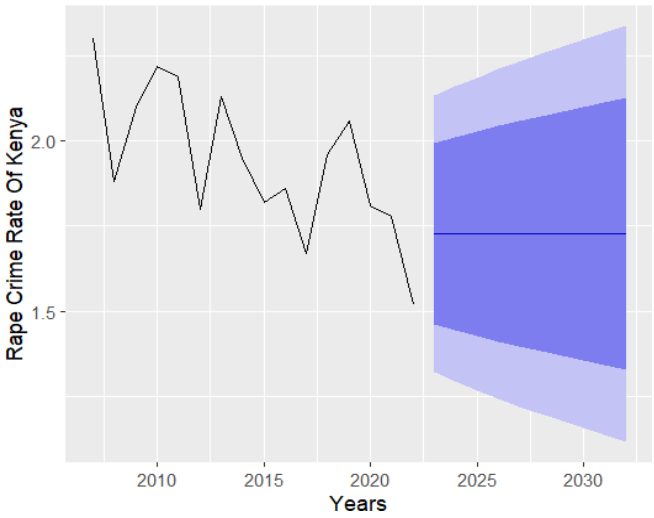
Now, for the said purpose we forecast the future time points by directly using the function auto. Arima () in R-programming which gives the best fitted model with MA (Moving Average), AR(Auto-Regressive) and differencing parameters, according to lowest AICc (Akaike Information Criterion Corrected) of the given data set.

Figure-21 (Time Series Plot)



From this above time series plot we can conclude that overall shape of the time series indicates a general downward trend in Kenya's rape crime rate with short-term fluctuations over the 16 years period.

Figure-22 (Forecasting Graph)
Forecast from Arima(0,1,1)



From the above forecasting plot, we can conclude that the data fits in the Arima (0,1,1) with non-zero mean model which is notated as;

$y_t = y_{t-1} + \mu + \epsilon_t + \theta\epsilon_{t-1}$, where y_t is the time series value, y_{t-1} is the previous value, μ is constant or drift term, $\epsilon_t \sim N(0, \sigma 2)$ is the white noise, θ MA (1) coefficient and ϵ_{t-1} MA(1) introduces short-term memory in errors.

Forecasted values:

Table: 22.1 showing the forecasted values for the next **10 year (2023-2022)** of **Kenya**

Years	Forecast Points
2023	1.727274

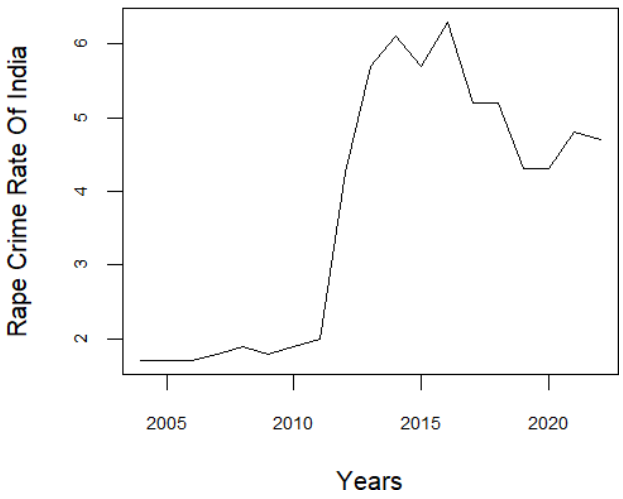
2024	1.727274
2025	1.727274
2026	1.727274
2027	1.727274
2028	1.727274
2029	1.727274
2030	1.727274
2031	1.727274
2032	1.727274

Rape crime rate in Kenya will likely stabilize modestly over the next several years (2023-2032) if we assume the current trends continue.

III. **Forecasting rape crime cases for India:** We will try to forecast the data of future values of rape cases based on all over India using the data of past 19 years. For forecasting future time points, we have considered the dataset corresponding to the rape crime cases for previous 19 years (2004-2022) for India and we are trying to predict the future values for the next 10 years (2023-2032).

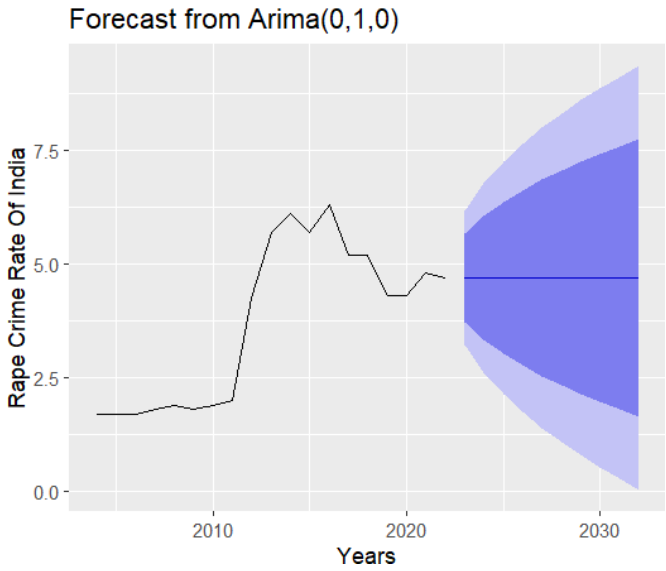
Now, for the said purpose we forecast the future time points by directly using the function auto. Arima () which gives the best fitted model with MA (Moving Average), AR(Auto-Regressive) and differencing parameters according to the lowest AICc (Akaike Information Criterion Corrected) of the given data set.

Figure-23 (Time Series Plot)



From this above time series plot we can conclude that the rape crime rate in India exhibits a transformational shift post-2012. Initially flat, it sharply rises then stabilizes with minor fluctuations. The current trend shows a gradual normalization at a higher level than the early 2000s or might be due covid-19's lockdown period from 2020-2022 years.

Figure-24 (Forecasting Graph)



From the above forecasting plot, we can conclude this model is Arima (0,1,0) which is;

$y_t = y_{t-1} + \epsilon_t$, where y_t is the original time series, y_{t-1} is the first difference of the time series after which it becomes stationary and ϵ_t is the white noise at time t.

Now, this model is a **pure random walk** that is; new values is just the previous value plus some random noises. Also, the model expects the future average rate to remain stable with no upwards or downwards shift.

Forecasted values:

Table: 24.1 showing the forecasted values for the next **10 year (2023-2022)** of **India**.

Years	Forecast Points
2023	4.7
2024	4.7
2025	4.7
2026	4.7
2027	4.7
2028	4.7
2029	4.7
2030	4.7
2031	4.7
2032	4.7

Rape crime rate in India will likely stabilize modestly over the next several years (2023-2032) if we assume the current trends continue.

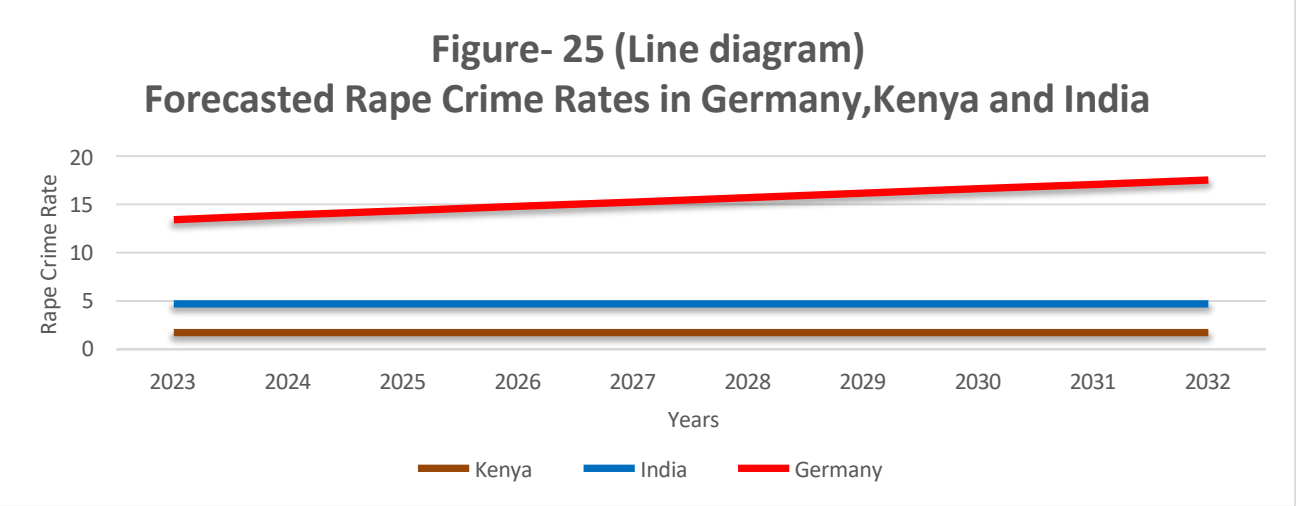
Comparison of the three countries:

We can compare among Germany, Kenya and India with respect to our forecasted future rape crime rate values for upcoming 10 years (2023-2022).

Table Of Comparison

Table-6, showing the **past years data points (2004-2022)** and **next upcoming years forecasted values (2023-2032)** of rape crime rate of Germany, Kenya and India.

Germany				Kenya				India			
Year	Rape crime rate	Year	Forecasted rape crime rate	Year	Rape crime rate	Year	Forecasted rape crime rate	Year	Rape crime rate	Year	Forecasted rape crime rate
2004	10.87	2023	13.41399	2004	4.14	2023	1.727274	2004	1.7	2023	4.7
2005	10.01	2024	13.91169	2005	3.81	2024	1.727274	2005	1.7	2024	4.7
2006	10	2025	14.33503	2006	3.5	2025	1.727274	2006	1.7	2025	4.7
2007	9.25	2026	14.81084	2007	2.3	2026	1.727274	2007	1.8	2026	4.7
2008	8.98	2027	15.24963	2008	1.88	2027	1.727274	2008	1.9	2027	4.7
2009	9	2028	15.71454	2009	2.1	2028	1.727274	2009	1.8	2028	4.7
2010	9.5	2029	16.16102	2010	2.22	2029	1.727274	2010	1.9	2029	4.7
2011	9.26	2030	16.62050	2011	2.19	2030	1.727274	2011	2	2030	4.7
2012	9.85	2031	17.07081	2012	1.8	2031	1.727274	2012	4.26	2031	4.7
2013	9.07	2032	17.52760	2013	2.13	2032	1.727274	2013	5.69	2032	4.7
2014	8.97			2014	1.95			2014	6.1		
2015	8.56			2015	1.82			2015	5.7		
2016	9.62			2016	1.86			2016	6.3		
2017	10.06			2017	1.67			2017	5.2		
2018	10.9			2018	1.96			2018	5.2		
2019	11.52			2019	2.06			2019	4.3		
2020	12.15			2020	1.81			2020	4.3		
2021	12.48			2021	1.78			2021	4.8		
2022	15.06			2022	1.52			2022	4.7		



From table no-6 and figure- 25, we can conclude that for upcoming 10 years the forecasted rape crime rate in India and Kenya shows a stabilised trend whereas, the forecasted rape crime rate in Germany shows an increasing trend.

Analysis and Model Fitting of the of 10 states in India

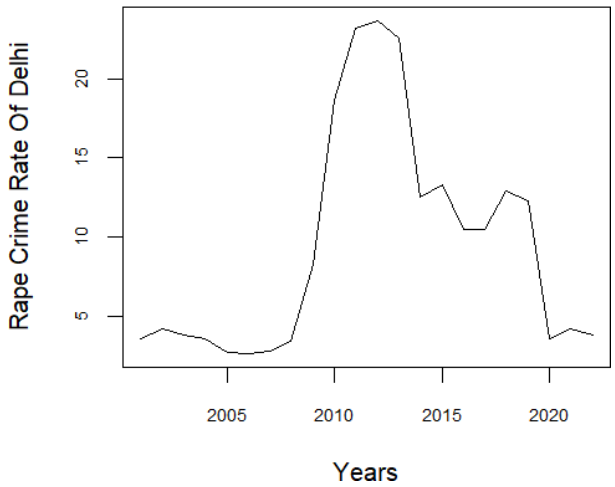
We are trying to forecast the dataset corresponding to the past 19 years of rape crime rate data across top 10 states of India for upcoming 10 years (2023-2032) and after getting the forecasted values on the basis of the forecasted values of top 10 states we can check if the ranking of the top 10 states remains the same over the next 10 years or not.

Now, we will try to forecast the data of future values of rape crime rate based on all 10 states separately using the past 19 years data. For forecasting the future time points, we have considered the dataset corresponding to the rape crimes cases for previous 19 years (2004 -2022) and predicting the future values for the next 10 years (2004-2022).

Now, for the said purpose we forecast the future time points by directly using the function auto. Arima () which gives the best fitted model with MA (Moving Average), AR(Auto-Regressive) and differencing parameters.

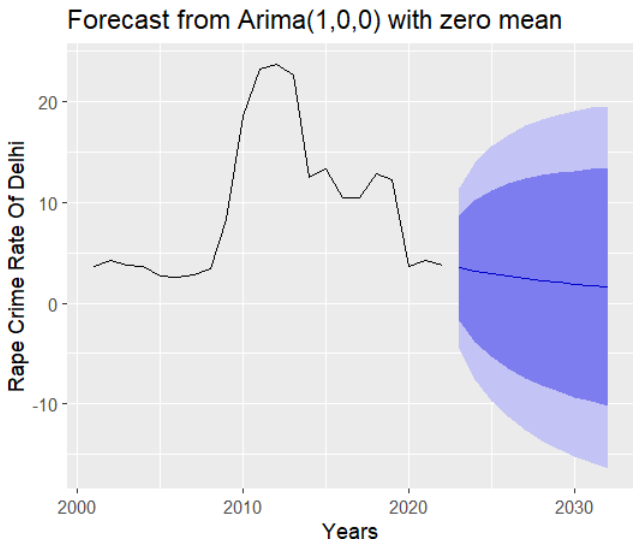
I. Delhi:

Figure-25 (Time Series Plot)



From figure- 25 time series plot we can conclude that the rape crime rate in Delhi has shown high volatility since the past years and after the Nirbhaya case which marked a clear inflection point from which cases decreased slightly. Meanwhile later we see short-term fluctuations in the cases.

Figure-26 (Forecasting Graph)



From figure- 26 forecasting graph, we can conclude that the data fits right in Arima (1,0,0) with non -zero mean which is notated as;

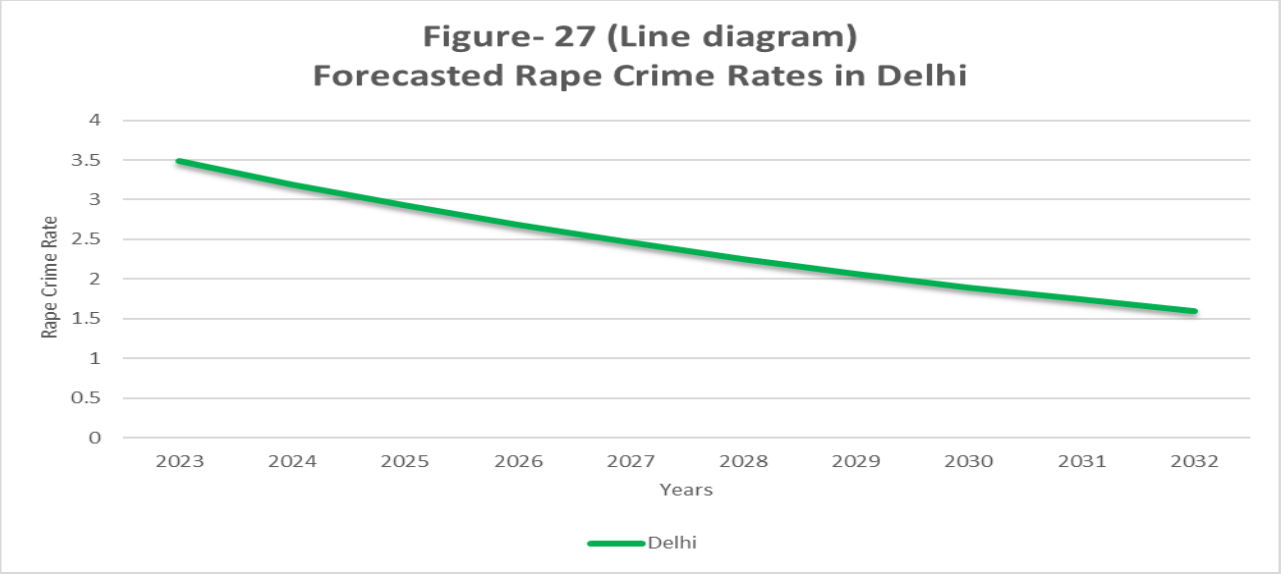
$y_t = \phi y_{t-1} + \epsilon_t$, where y_t is the time series value at time t, ϕ is the AR (auto-regressive) coefficient, y_{t-1} is the previous value and $\epsilon_t \sim N(0, \sigma 2)$ is white noise.

The moving average component with order 0 means that no past forecast errors are included in the prediction. The model would estimate a coefficient that represents the strength of the relationship between previous year's cases and next year's cases, as well as a constant term and a random error component.

Forecasted values:

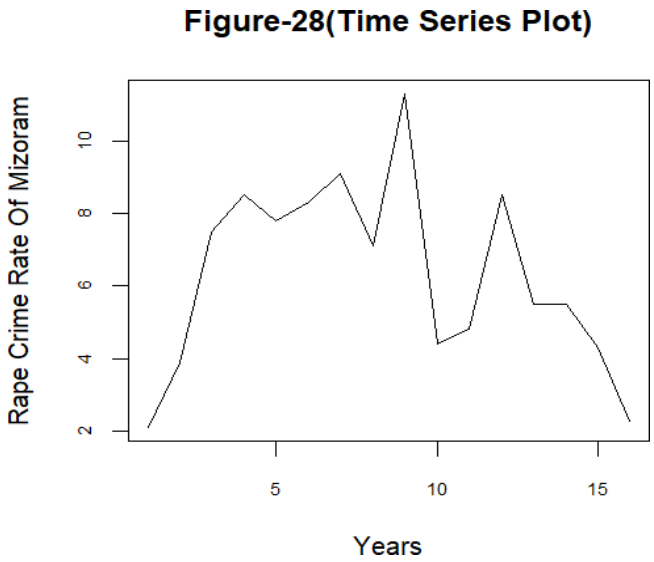
Table: 26.1 showing the forecasted values for the next **10 year (2023-2022)** of **Delhi**.

Years	Forecasted Points
2023	3.484206
2024	3.194656
2025	2.929168
2026	2.685744
2027	2.462549
2028	2.257902
2029	2.070262
2030	1.898216
2031	1.740467
2032	1.595828



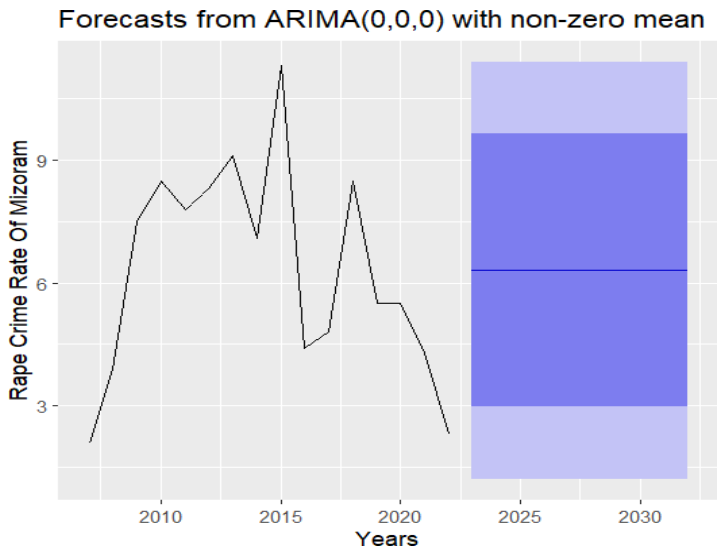
From figure- 27 and table no- 26.1, we can conclude that in beginning of the next forecasted years there is an increase but the crime rate gradually decreases for the next years. Hence shows a decline.

II. Mizoram:



From figure- 28 time series plot, we can conclude that the rape crime rate in Mizoram has short term fluctuations over the years (2004-2022) and maximum number of cases had occurred in the year 2012 later declined by 2013 and minimum cases got reported in 2020.

Figure-29 (Forecasting Graph)



From figure- 29 forecasting graph, we can conclude that the data fits right in Arima (0,0,0) with non-zero mean which is notated as;

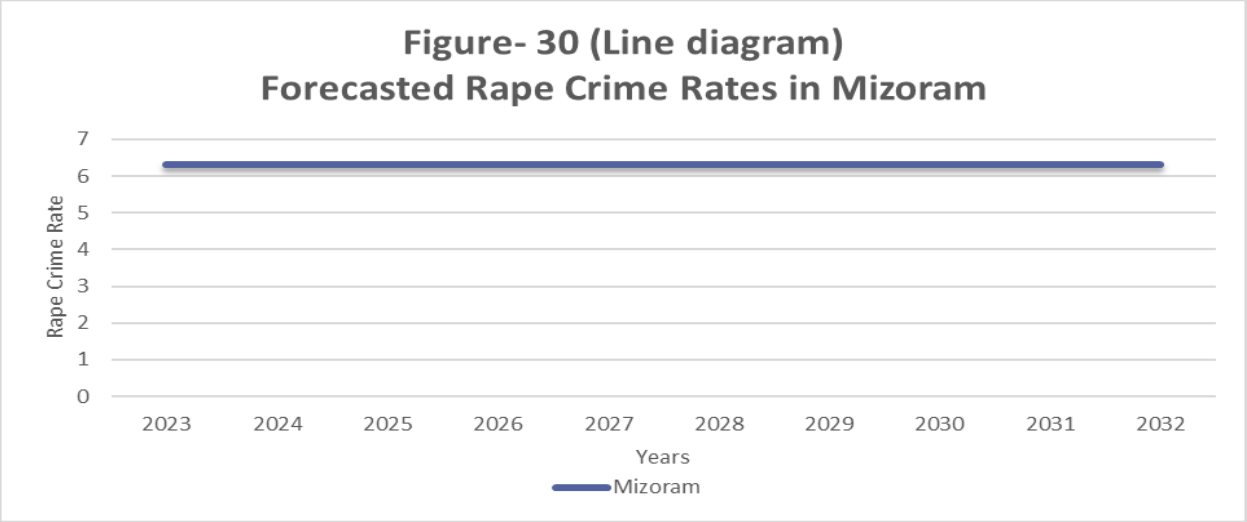
$y_t = \mu + \epsilon_t$, where y_t is the time series value at time t, μ is constant mean (non-zero as specified) and ϵ_t is random white noise.

In this model the data fluctuates randomly around a constant mean value. It is the simplest time series model possible where the future looks like the average of the past, plus random noise.

Forecasted values:

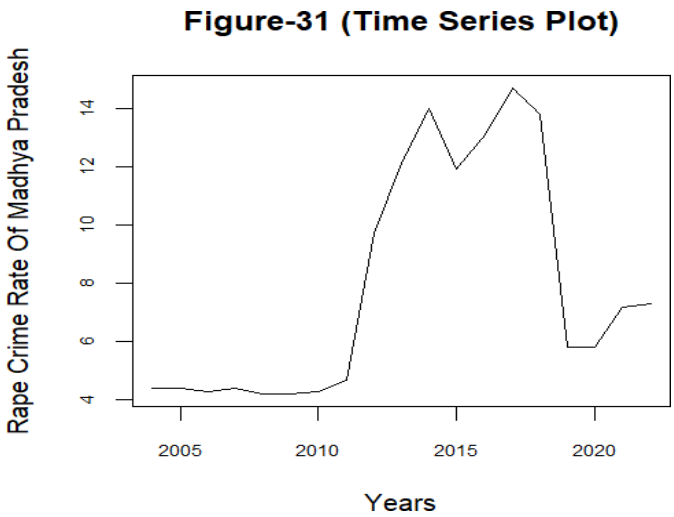
Table: 29.1 showing the forecasted values for the next **10 year (2023-2022)** of **Mizoram**.

Years	Forecast Points
2023	6.30625
2024	6.30625
2025	6.30625
2026	6.30625
2027	6.30625
2028	6.30625
2029	6.30625
2030	6.30625
2031	6.30625
2032	6.30625



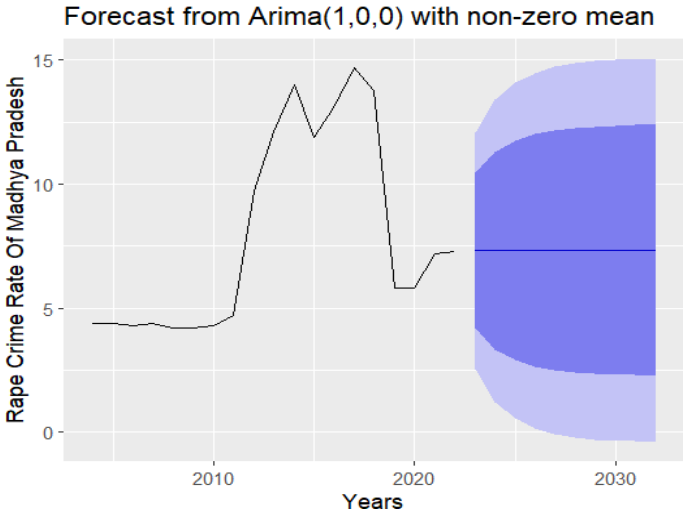
From figure- 30 and table no- 29.1, we can conclude that for the next 10 years the rape crime rate in Mizoram stabilizes at 6.3 if the current trend remains the same.

III. Madhya Pradesh:



From figure- 31 time series plot, we can conclude that the rape crime rate in Madhya Pradesh has lower cases reported in the beginning years, then peak crime rate in the years 2012-2013 and lowered again in the year 2020 due to COVID-19 pandemic. Later post pandemic again shows an increase in cases.

Figure-32 (Forecasting Graph)



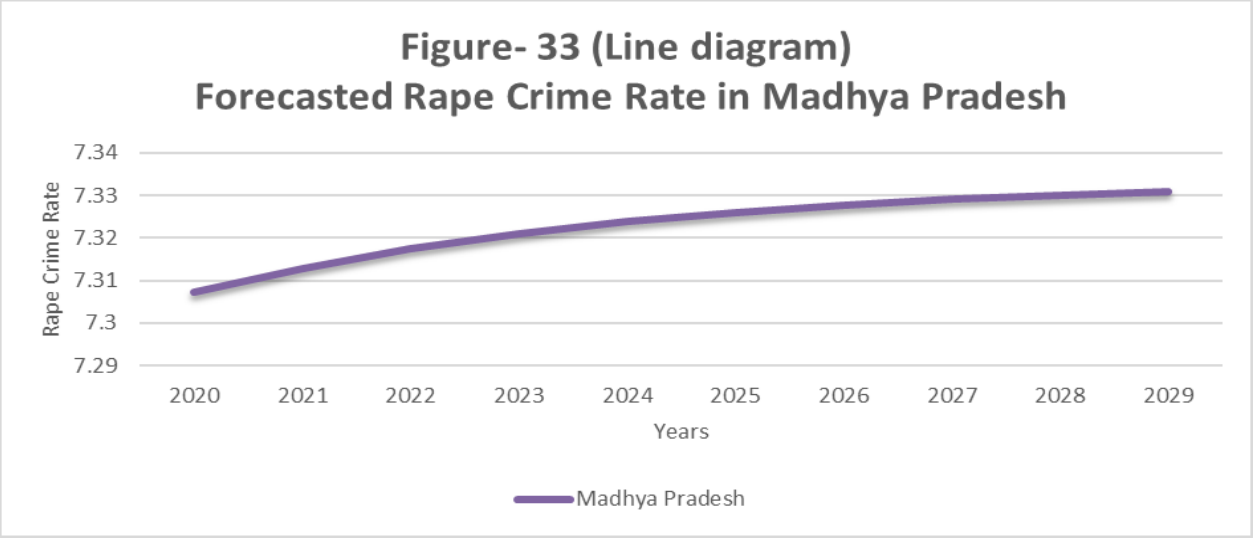
From figure- 32 forecasting graph, we can conclude that the data fits right in Arima (1,0,0) with non -zero mean which is notated as;

$y_t = \mu + \phi y_{t-1} + \epsilon_t$, where y_t is the time series value at time t, μ is a constant or drift term, ϕ is the AR (auto-regressive) coefficient, y_{t-1} is the previous value and $\epsilon_t \sim N(0, \sigma 2)$ is white noise. The moving average component with order 0 means that no past forecast errors are included in the prediction.
The model would estimate a coefficient that represents the strength of the relationship between previous year's cases and next year's cases, as well as a constant term and a random error component.

Forecasted values:

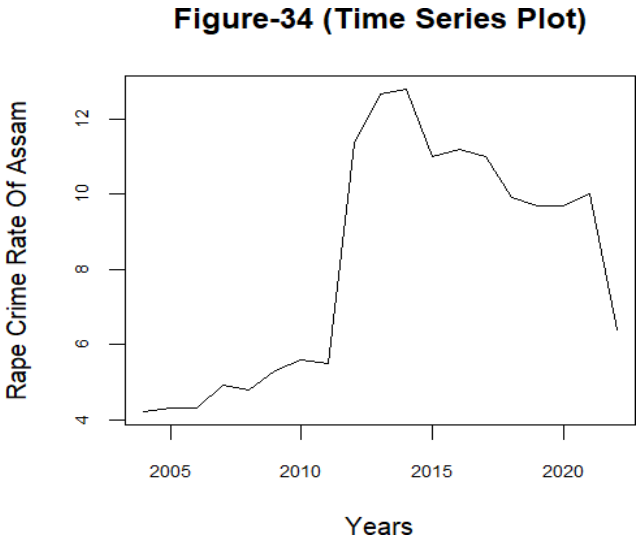
Table: 32.1 showing the forecasted values for the next **10 year (2023-2022)** of **Madhya Pradesh**.

Year	Point Forecast
2023	7.307249
2024	7.312956
2025	7.317449
2026	7.320986
2027	7.323771
2028	7.325963
2029	7.327688
2030	7.329047
2031	7.330116
2032	7.330958



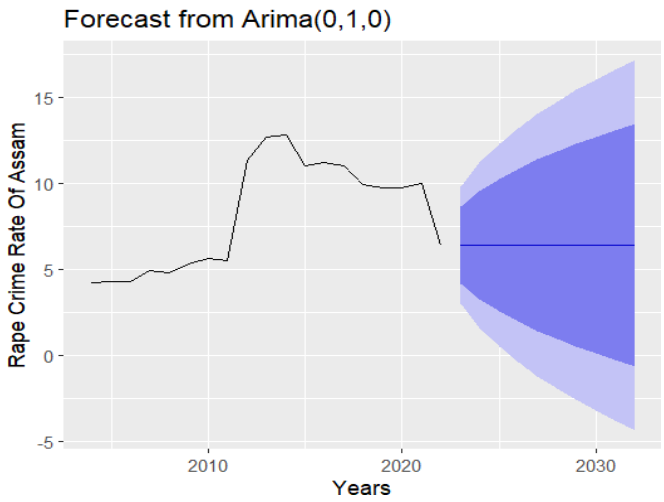
From figure- 33 and table no- 32.1, we can conclude that in beginning of the next forecasted years there is a decrease but in the later years it's seen to increase by point 31 for the next 9 years (2024-2032).

IV. Assam:



From figure- 34 time series plot, we can conclude that the rape crime rate in Assam has peak crime rate in the years 2012-2013 with short-term fluctuations. Like other states during COVID-19 pandemic, the cases during 2020 didn't go down despite lockdown all over India.

Figure-35 (Forecasting Graph)



From figure- 35 forecasting graph, we can conclude that the data fits right in the model Arima (0,1,0) which is;

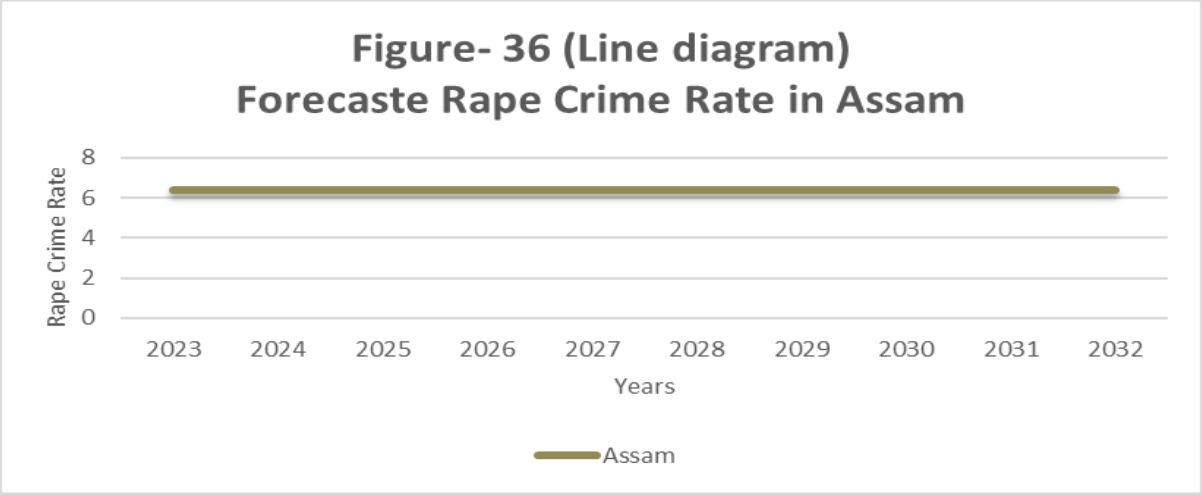
$y_t = y_{t-1} + \epsilon_t$, where y_t is the original time series, y_{t-1} is the first difference of the time series after which it becomes stationary and ϵ_t is the white noise at time t.

Now, this model is a **pure random walk** that is; new values is just the previous value plus some random noises. Also, the model expects the future average rate to remain stable with no upwards or downwards shift.

Forecasted values:

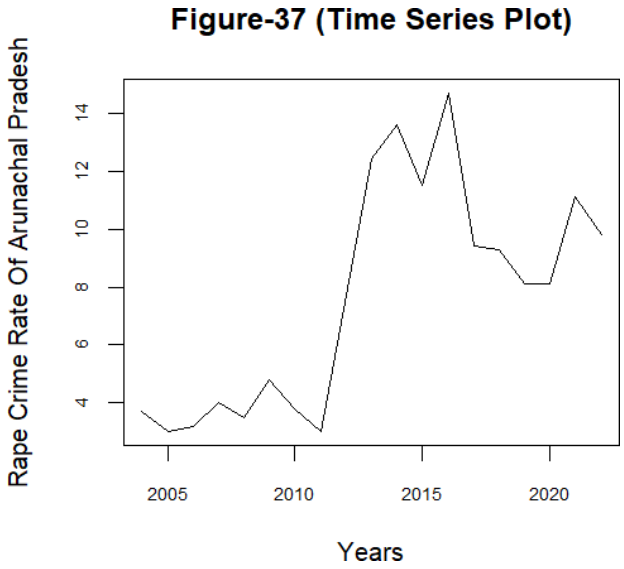
Table: 35.1 showing the forecasted values for the next **10 year (2023-2022)** of **Assam**.

Years	Forecast Points
2023	6.4
2024	6.4
2025	6.4
2026	6.4
2027	6.4
2028	6.4
2029	6.4
2030	6.4
2031	6.4
2032	6.4



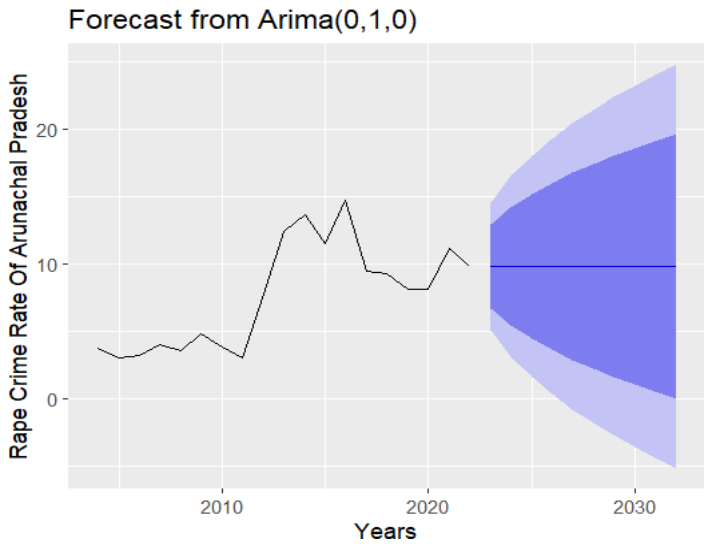
From figure- 36 and table no- 35.1, we can conclude that for the next 10 years the rape crime rate in Assam stabilizes at 6.4 if the current trend remains the same.

V. Arunachal Pradesh:



From figure- 37 time series plot, we can conclude that the rape crime rate in Arunachal Pradesh has short-term fluctuations with highest cases reported in the year 2015 and eventually decreased in 2020 due to COVID-19 pandemic but after withdrawing of first lockdown for few months in 2021 we could again see rise in cases.

Figure-38 (Forecasting Graph)



From figure- 38 forecasting graph, we can conclude that the data fits right in the model Arima (0,1,0) which is;

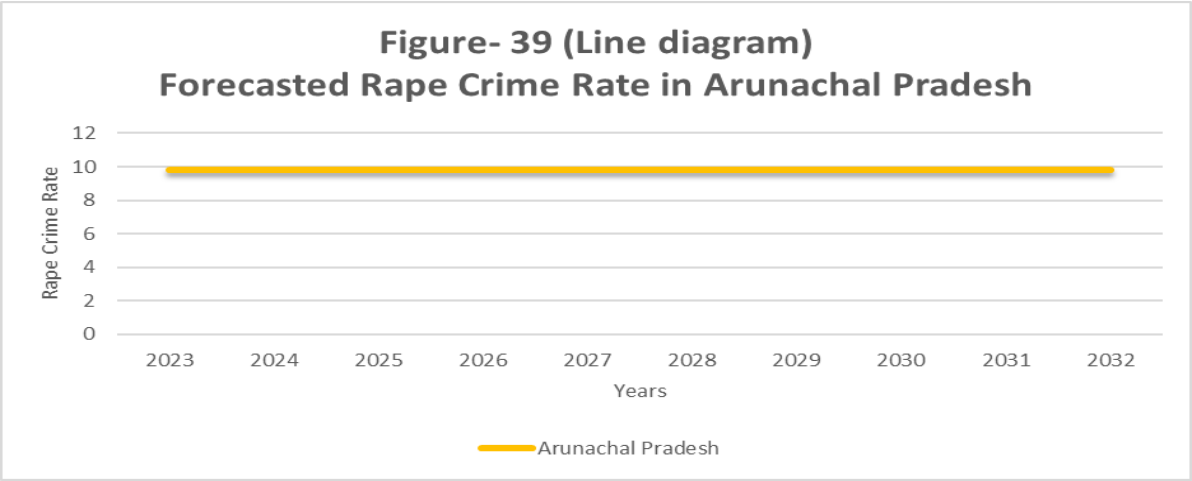
$y_t = y_{t-1} + \epsilon_t$, where y_t is the original time series, y_{t-1} is the first difference of the time series after which it becomes stationary and ϵ_t is the white noise at time t.

Now, this model is a **pure random walk** that is; new values is just the previous value plus some random noises. Also, the model expects the future average rate to remain stable with no upwards or downwards shift.

Forecasted values:

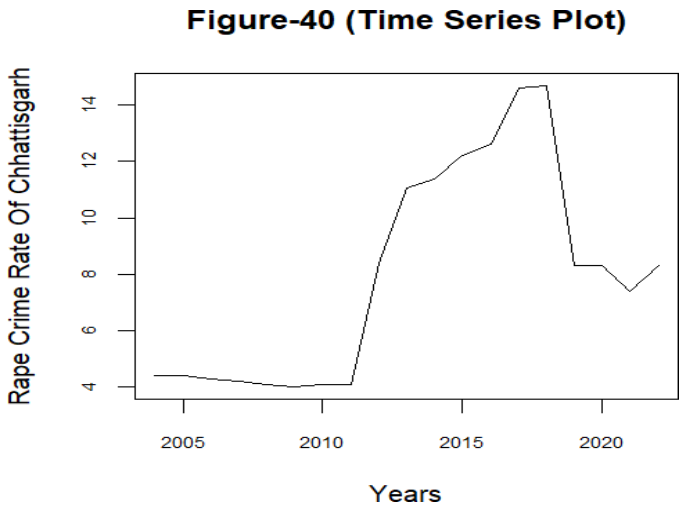
Table: 38.1 showing the forecasted values for the next **10 year (2023-2022)** of **Arunachal Pradesh**.

Years	Forecast Points
2023	9.8
2024	9.8
2025	9.8
2026	9.8
2027	9.8
2028	9.8
2029	9.8
2030	9.8
2031	9.8
2032	9.8



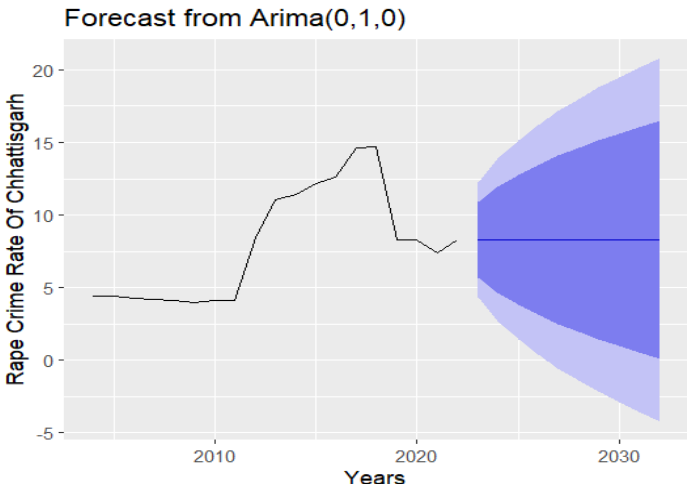
From figure- 39 and table no- 38.1, we can conclude that for the next 10 years the rape crime rate in Arunachal Pradesh stabilizes at 9.8 if the current trend remains the same.

VI. Chhattisgarh:



From figure- 40 time series plot, we can conclude that the rape crime rate in Chhattisgarh has steady or rather lower cases reported from 2005-2010 years, the year 2017 had peaked in the reported number of cases and eventually decreased in 2020-2021 due to COVID-19 pandemic.

Figure-41 (Forecasting Graph)



From figure- 42 forecasting graph, we can conclude that the data fits right in the model Arima (0,1,0) which is;

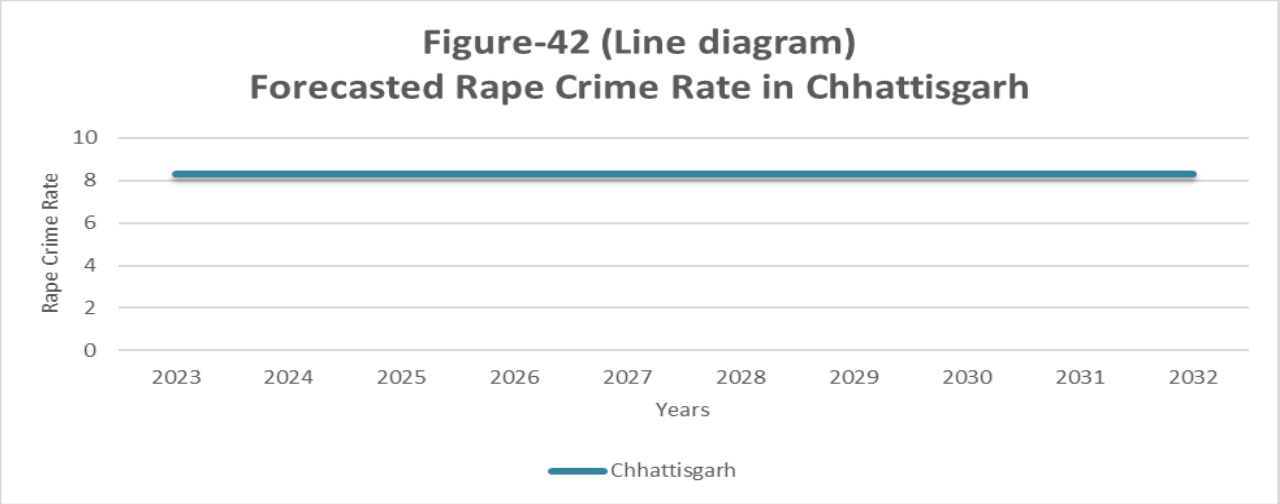
$y_t = y_{t-1} + \epsilon_t$, where y_t is the original time series, y_{t-1} is the first difference of the time series after which it becomes stationary and ϵ_t is the white noise at time t.

Now, this model is a **pure random walk** that is; new values is just the previous value plus some random noises. Also, the model expects the future average rate to remain stable with no upwards or downwards shift.

Forecasted values:

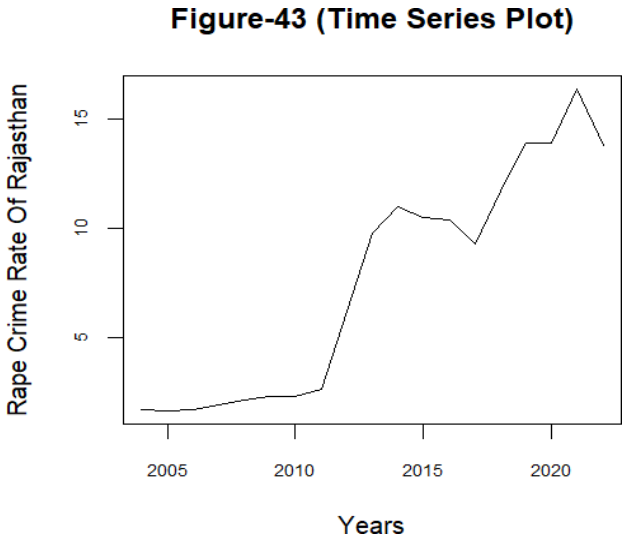
Table: 41.1 showing the forecasted values for the next **10 year (2023-2022)** of **Chhattisgarh**.

Years	Forecasted Points
2023	8.3
2024	8.3
2025	8.3
2026	8.3
2027	8.3
2028	8.3
2029	8.3
2030	8.3
2031	8.3
2032	8.3



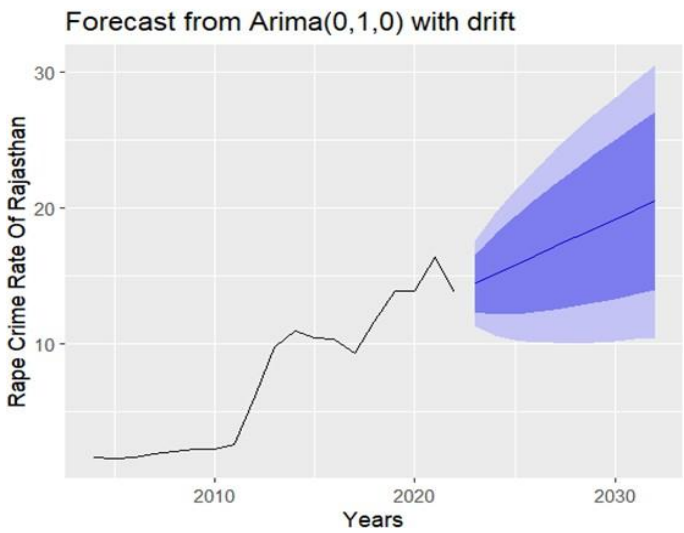
From figure- 42 and table- 41.1, we can conclude that for the next 10 years the rape crime rate in Chhattisgarh stabilizes at 8.3 if the current trend remains the same.

VII. Rajasthan:



From figure- 43 time series plot, we can conclude that the rape crime rate in Rajasthan has steady or rather lower cases reported from 2005-2010 years, then increased in the year 2014. Later, highest cases were reported in the year 2021 eventually due the withdraw of the first lockdown for few months.

Figure-44 (Forecasting Graph)



From figure- 44 forecasting graph, we can conclude that the data fits right in the model Arima (0,1,0) with drift which is;

$y_t = \mu + y_{t-1} + \epsilon_t$, where y_t is the original time series, μ is the drift or constant, y_{t-1} is the first difference of the time series after which it becomes stationary and ϵ_t is the white noise at time t.

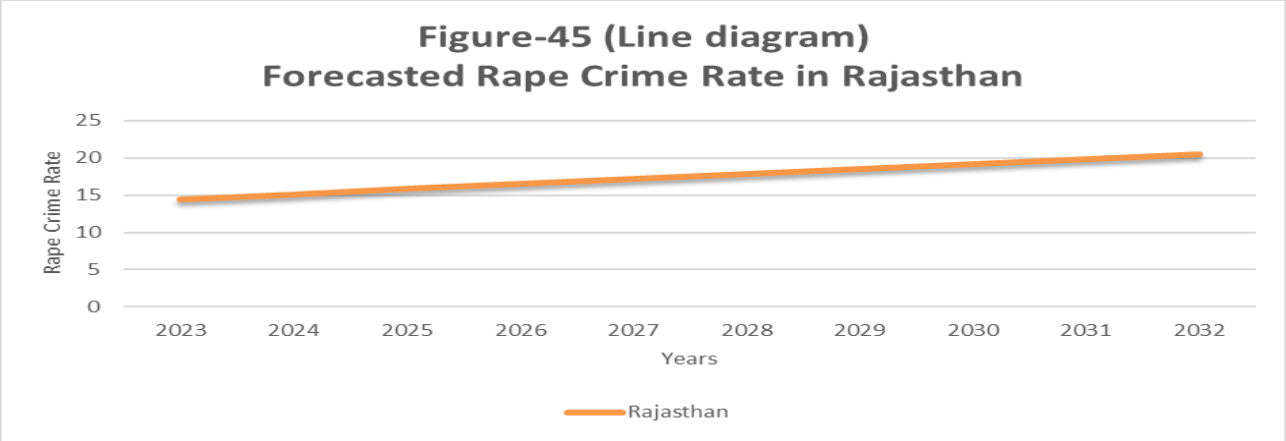
Now, fitting the drift coefficient in the model we get;

$y_t = 0.6722 + y_{t-1} + \epsilon_t$

Forecasted values:

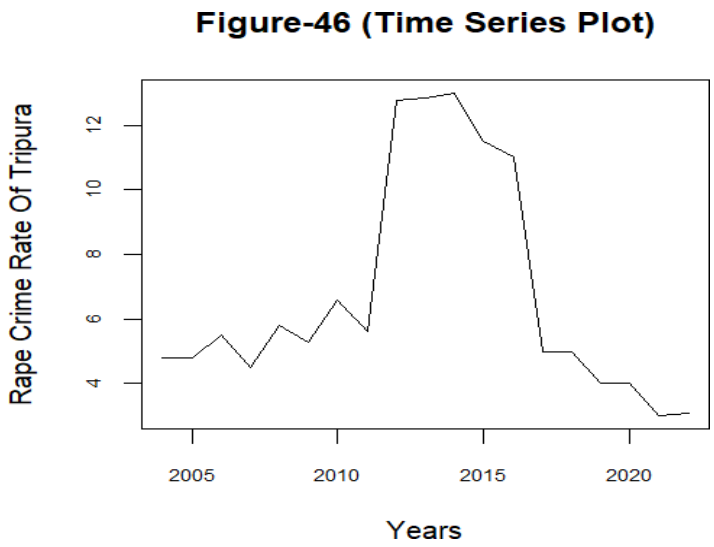
Table: 44.1 showing the forecasted values for the next **10 year (2023-2022)** of **Rajasthan**.

Years	Forecasted Points
2023	14.47222
2024	15.14444
2025	15.81667
2026	16.48889
2027	17.16111
2028	17.83333
2029	18.50556
2030	19.17778
2031	19.85
2032	20.52222



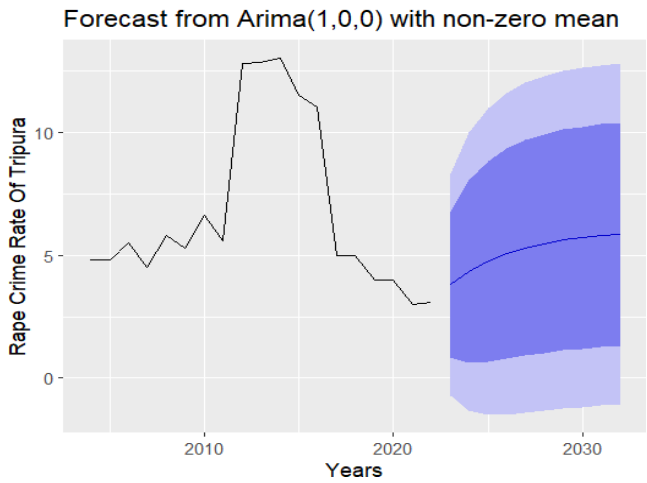
From figure 45 and table- 44.1, we can conclude that in the beginning the crime rates are low but seems to increase over the years, showing an upward trend.

VIII. Tripura:



From figure- 46 time series plot, we can conclude that the rape crime rate in Tripura has short-term fluctuations towards the beginning years, then increased in the years 2012-2014. Later, decreased drastically for the next years till 2021 with slight increase from the year 2022.

Figure-47 (Forecasting Graph)



From figure- 47 forecasting graph, we can conclude that the data fits right in Arima (1,0,0) with non-zero mean which is notated as;

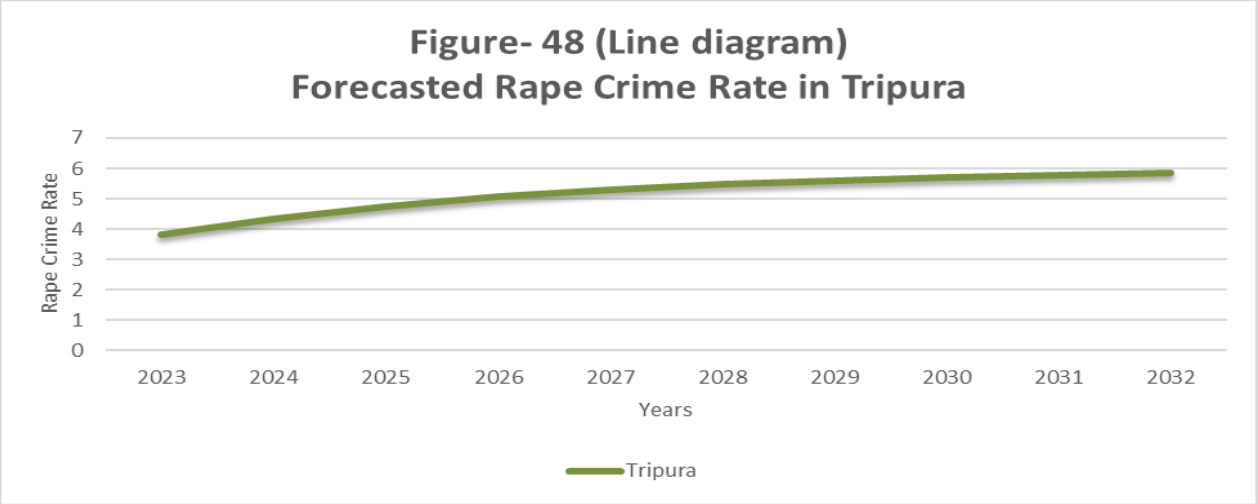
$y_t = \mu + \phi y_{t-1} + \epsilon_t$, where y_t is the time series value at time t, μ is a constant or drift term, ϕ is the AR (auto-regressive) coefficient, y_{t-1} is the previous value and $\epsilon_t \sim N(0, \sigma 2)$ is white noise. The moving average component with order 0 means that no past forecast errors are included in the prediction.

The model would estimate a coefficient that represents the strength of the relationship between previous year's cases and next year's cases, as well as a constant term and a random error component.

Forecasted values:

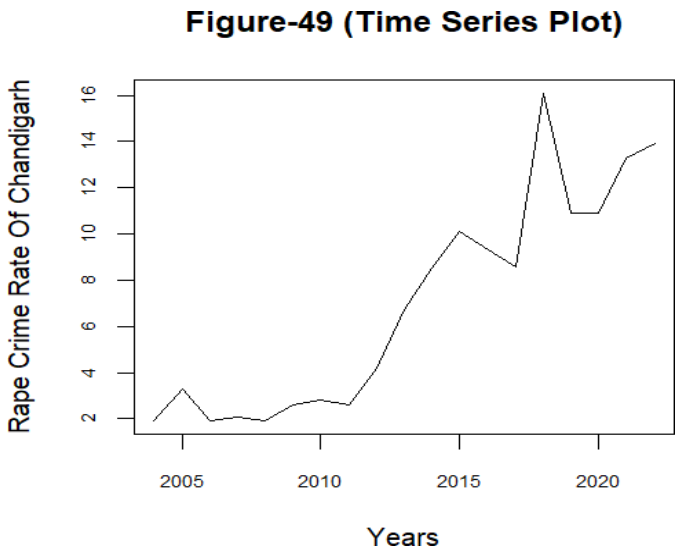
Table: 47.1 showing the forecasted values for the next **10 year (2023-2022)** of **Tripura**.

Years	Forecasted Points
2023	3.803531
2024	4.338823
2025	4.746108
2026	5.055996
2027	5.291779
2028	5.471178
2029	5.607677
2030	5.711534
2031	5.790555
2032	5.850679



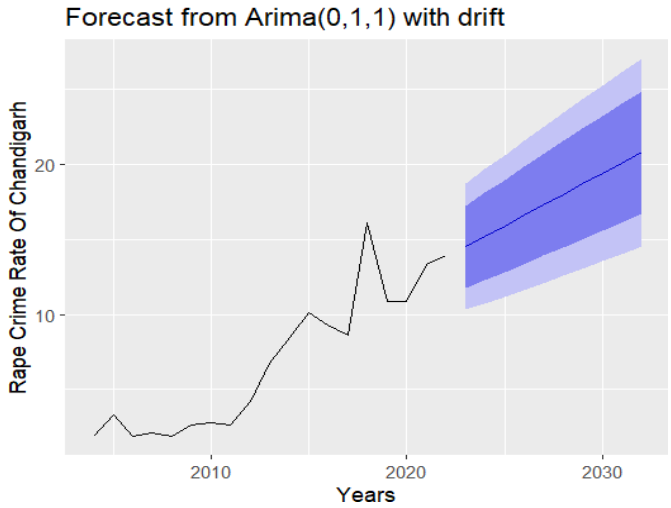
From figure- 48 and table no- 47.1, we can conclude that in the beginning the crime rates are low but seems to increase over the years, showing an upward trend.

IX. Chandigarh:



From figure- 49 time series plot, we can conclude that the rape crime rate in Chandigarh has short-term fluctuations towards the beginning years, then experienced the highest reported cases in the year 2018. Later, decreased due to COVID-19 pandemic in the year 2020 because of lockdown all over India but unfortunately increased in the year 2021 and continued to increase.

Figure-50 (Forecasting Graph)



From figure- 50 forecasting graph, we can conclude that the data fits in the Arima (0,1,1) with drift which is notated as;

$y_t = y_{t-1} + \mu + \epsilon_t + \theta_1 (\epsilon_{t-1})$, where y_t is the time series value, y_{t-1} is the previous value, μ is constant or drift term, $\epsilon_t \sim N(0, \sigma 2)$ is the white noise, θ_1 is the MA (1) coefficient with impact of the previous period's error on the current value and ϵ_{t-1} MA (1) introduces short-term memory in errors.

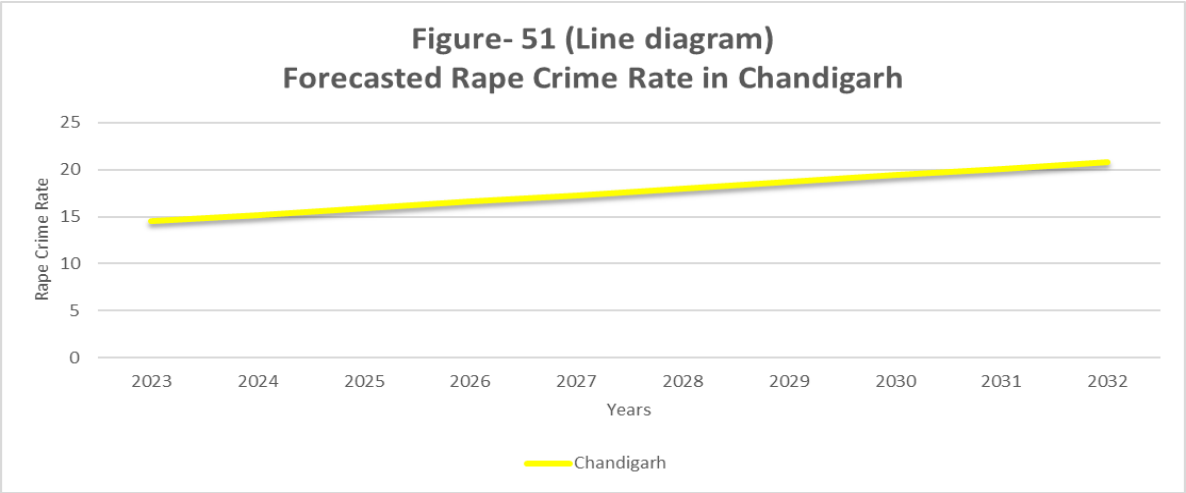
Now, fitting the drift coefficient in the model we get;

$y_t = 0.6984 + \epsilon_t + \theta_1 (\epsilon_{t-1})$

Forecasted values:

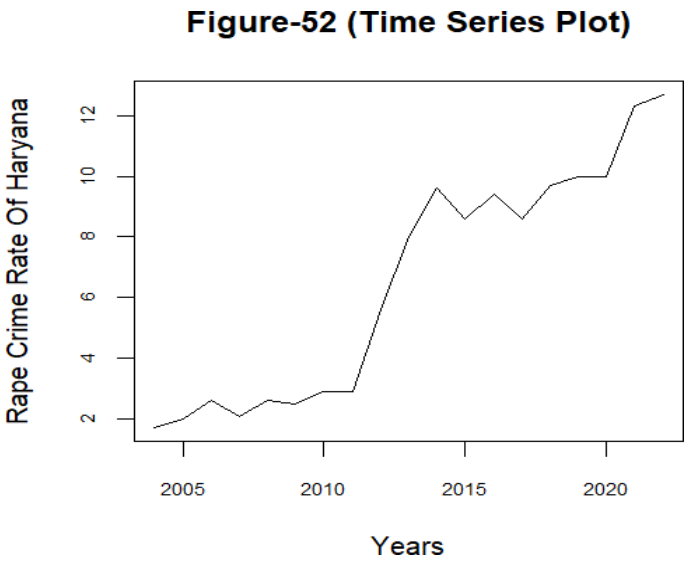
Table: 50.1 showing the forecasted values for the next **10 year (2023-2022)** of **Chandigarh**.

Years	Forecasted Points
2023	14.50423
2024	15.20266
2025	15.9011
2026	16.59953
2027	17.29796
2028	17.9964
2029	18.69483
2030	19.39326
2031	20.0917
2032	20.79013



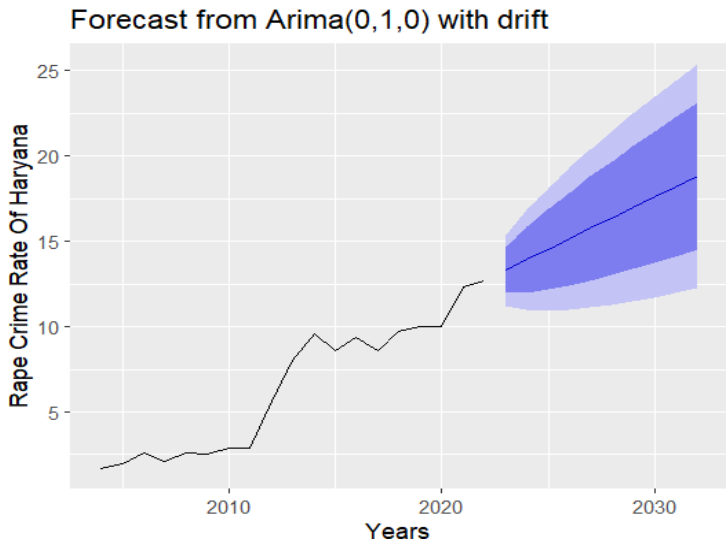
From figure- 51 and table no- 50.1, we can conclude that in the beginning the crime rates will be low but seems to increase over the years, showing an upward trend.

X. Haryana:



From figure- 52 time series plot, we can conclude that the rape crime rate in Haryana has short-term fluctuations, then experienced an increase in the reported cases for the year 2012. Later, in spite of lockdown in the years 2020-2022 due to COVID-19 pandemic this state saw no decrease in the rape crime rate and rather saw its highest increase in the year 2021.

Figure-53 (Forecasting Graph)



From figure- 53 forecasting graph, we can conclude that the data fits right in the model Arima (0,1,0) with drift which is;

$y_t = \mu + y_{t-1} + \epsilon_t$, where y_t is the original time series, μ is the drift or constant, y_{t-1} is the first difference of the time series after which it becomes stationary and ϵ_t is the white noise at time t.

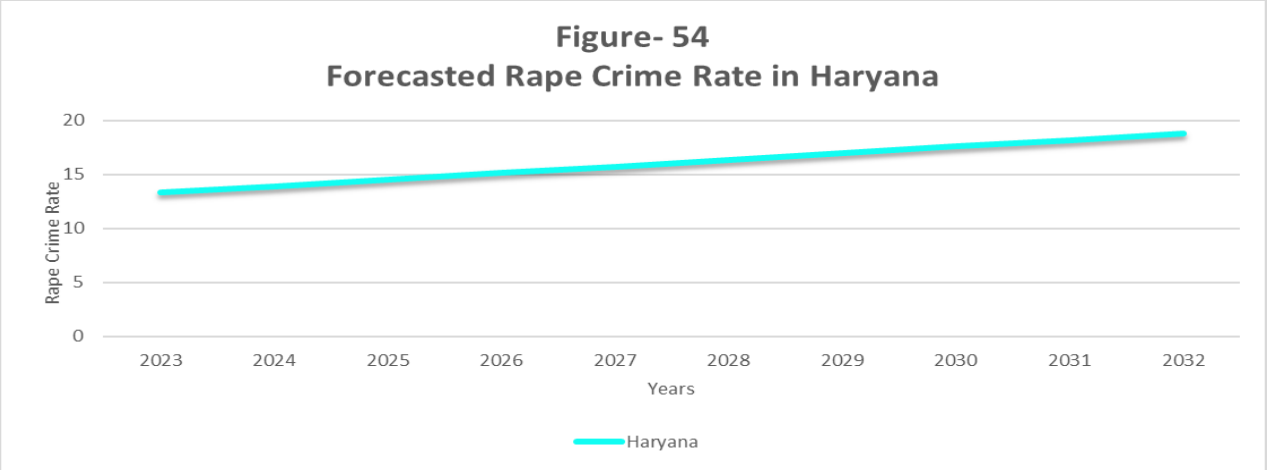
Now, fitting the drift coefficient in the model we get;

$y_t = 0.6111 + y_{t-1} + \epsilon_t$

Forecasted values:

Table: 53.1 showing the forecasted values for the next **10 year (2023-2022)** of **Haryana**.

Years	Forecasted Points
2023	13.31111
2024	13.92222
2025	14.53333
2026	15.14444
2027	15.75556
2028	16.36667
2029	16.97778
2030	17.58889
2031	18.2
2032	18.81111



From figure- 54 and table no-53.1, we can conclude that in the beginning the crime rates are low but seems to increase over the years, showing an upward trend.

Future ranking of the states based on the forecasted values

Ranking the top 10 states which were heading the top 10 title on the rape crime rate from the year 2004-2022 for the past 19 years and now checking the top 10 states for the next 10 years 2023-2032 based on their forecasted values.

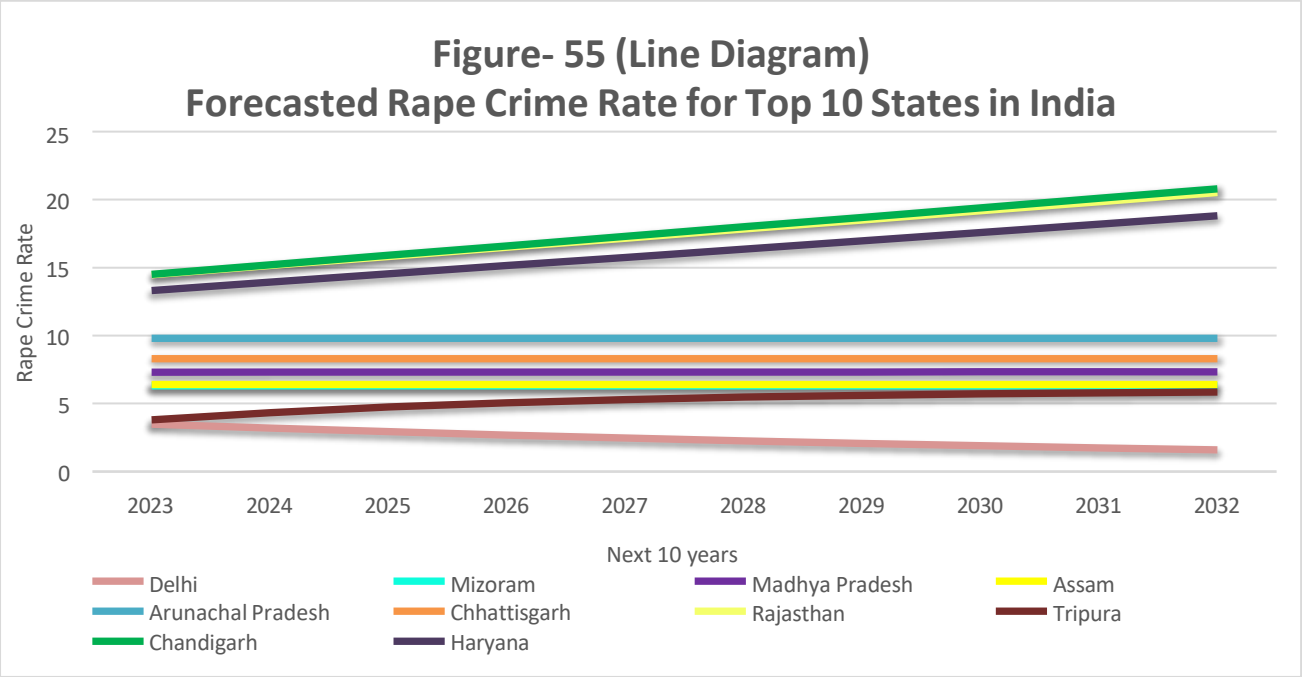


Table For Ranking

Table- 7, showing the predicted rank of the past **top 10 states** for the **next upcoming 10 years**.

Rank	States	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Chandigarh	14.50423	15.20266	15.9011	16.59953	17.29796	17.9964	18.69483	19.39326	20.0917	20.79013
2	Rajasthan	14.47222	15.14444	15.81667	16.48889	17.16111	17.83333	18.50556	19.17778	19.85	20.52222
3	Haryana	13.31111	13.92222	14.53333	15.14444	15.75556	16.36667	16.97778	17.58889	18.2	18.81111
4	Arunachal Pradesh	9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.8
5	Chhattisgarh	8.3	8.3	8.3	8.3	8.3	8.3	8.3	8.3	8.3	8.3
6	Madhya Pradesh	7.307249	7.312956	7.317449	7.320986	7.323771	7.325963	7.327688	7.329047	7.330116	7.330958
7	Assam	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4
8	Mizoram	6.30625	6.30625	6.30625	6.30625	6.30625	6.30625	6.30625	6.30625	6.30625	6.30625
9	Tripura	3.803531	4.338823	4.746108	5.055996	5.291779	5.471178	5.607677	5.711534	5.790555	5.850679
10	Delhi	3.484206	3.194656	2.929168	2.685744	2.462549	2.257902	2.070262	1.898216	1.740467	1.595828

Hence, we can conclude that our **TOP 10** states of India for the **next 10 years** corresponding to the forecasted rape crime rate are Chandigarh, Rajasthan, Haryana, Arunachal Pradesh, Chhattisgarh, Madhya Pradesh, Assam, Mizoram, Tripura and Delhi.

Conclusion

Living in the 21st century and through statistical analysis we observe that a heinous crime like rape still prevails against women globally. Through comparative analysis of rape crime rates between India and other global regions, reveal significant variations in crime rates. We observe that a developed country like Germany might saw an increase in crime rate in the nearest future whereas, India and Kenya saw a stable growth for the upcoming years. Through forecast models such as ARIMA which indicates without structural societal changes.

Now moving to states-wise statistical analysis, we conclude that:

- For the state Delhi, we have Arima model: Arima (1,0,0) with zero mean which is the best fitted model for the given dataset.
- For the states Tripura and Madhya Pradesh, we have Arima model: Arima (1,0,0) with non-zero mean which is the best fitted model for the given dataset.
- For the states Assam, Arunachal Pradesh and Chhattisgarh we have Arima model: Arima (1,0,0) which is the best fitted model for the given dataset.
- For the states Rajasthan and Haryana, we have Arima model: Arima (1,0,0) which is the best fitted model for the given dataset.
- For the state Mizoram, we have Arima model: Arima (0,0,0) with non-zero mean which is the best fitted model for the given dataset.
- For the state Chandigarh, we have Arima model: Arima (0,0,1) with drift which is the best fitted model for the given dataset.

After comparison of the forecasted values, we get to conclude that Germany which previously had high rape crime rate, cases continue to increase for the upcoming years, whereas for India and Kenya cases show a stabilized trend for the upcoming next ten years. Within, India based on the forecasted values for Delhi and Mizoram we observe high decline in the rape crime rate for the upcoming next ten years, whereas in case of Chandigarh, Rajasthan and Haryana we observe high increase in the rape crime rate for the upcoming next ten years.

Through demographical analysis, though Madhya Pradesh, Chhattisgarh, and Haryana have large populations and the risk faced by women (likely about safety, violence, or discrimination) is still very high. Even, Uttar Pradesh has a large number of rape cases in total but when we adjust for its huge population size (per lakh population), the rate remains moderate due to its vast population. In that case Delhi and Mizoram saw increase in the cases per lakhs population for the years 2004-2022 in spite of having lesser area and population. So, if we only look at the total numbers without considering how big the population is, we might get the wrong idea about how bad the situation really is. Sometimes it may also happen cases don’t get reported due to social-stigma, victim blaming and gender inequality.

Ultimately, we can come to a conclusion that no matter how large the area is or how big the population of that area is people who commit rape do not belong to a particular region or society, they commit it out of their own pleasure without thinking about the consequences of the crime. If focusing on the literacy rate which plays a vital role then according to 2011 census Indian states having high literacy rates have highest rape rates too, for example like Delhi and Mizoram the two top states of India which had the highest rape crime rate for the years 2004-2022 had male literacy rate about 90.0 and 93.3 respectively. Hence, offenders who commit this crime need not to be uneducated as evidently educated people also commit this heinous crime. Moreover, psychologically offenders of this crime often have bad past traumas which instigate them to commit such crimes and families should look after the upbringing of their child in a more sincere manner. To further prevention of rape against women, government should pass more strict laws against the offenders and proper investigation should be done so that in the cases where people fake report and mislead could be caught and innocent people would not get punished. In foreign countries like USA implementations of survivor-centred reforms, comprehensive education programs, and strong gender equality policies have achieved measurable reductions in both incidence and rates. So, the faster we take action the world becomes a better place for living.

Limitations & Future works

I. Data Availability and Reliability: This study is heavily dependent on secondary data from government and international sources. In many countries, underreporting of rape cases due to stigma, fear, or lack of legal support affects the reliability of available statistics.
II. Differences in Legal Definitions: The definition of rape and sexual assault varies significantly across different countries and jurisdictions. This inconsistency can lead to challenges in making direct comparisons.
III. Sample Size: In this project sample size was a problem as to be a perfect fit for Arima model we minimum require a sample size of 50-100.
IV. Statistical Model Limitations: Being a B.sc student my knowledge was limited and also because of time and resource constraints, only basic comparative techniques, time series model and graphical representations were used, rather than more sophisticated model approaches.

While this study primarily uses descriptive statistics and time series analysis future research can apply more advanced statistical methods. Techniques such as multivariate regression analysis, time series forecasting and cluster analysis can help uncover hidden patterns and trends in rape statistics across different countries and regions. Using inferential statistics, we could also determine whether observed differences between countries are statistically significant rather than due to random variation. Further this research could explore how socio-economic factors—such as literacy rate, poverty levels, gender equality indices, urbanization, and unemployment—correlate with the incidence of rape. Regional analyses within countries, particularly in large nations like India, could highlight local disparities.

Appendix

R CODES:

I. **Code for Germany:**

```
GD=read.csv("C:/Users/Dell/OneDrive/Desktop/new/GD.csv",header=TRUE)
GD
colnames(GD)
summary(GD)
boxplot(GD$Germany,main="Figure-6
BOXPLOT FOR GERMANY",ylab="Rape Crime Rate",col="yellow")
outliersg_rm<-GD$Germany[!GD$Germany%in% boxplot.stats(GD$Germany)$out]
length(GD$Germany)
length(outliersg_rm)
length(GD$Germany)-length(outliersg_rm)
boxplot(outliersg_rm,main="Figure-6.1
BOXPLOT FOR GERMANY",ylab="Rape Crime Rate",col="yellow")
library(tseries)
library(forecast)
library(generics)
gd_ts<-ts(outliersg_rm,start=2004,frequency=1)
gd_ts
plot(gd_ts,main="Figure-19 (Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of
Germany")
myarima=auto.arima(gd_ts,trace=T,stepwise=F,approximation=F)
accuracy(myarima)
par(cex.axis=0.7)
autoarima1<-
autoplot(forecast(auto.arima(gd_ts,trace=T,stepwise=F,approximation=F)),main="Figure-20
(Forecasting Graph)
Forecast from Arima(1,2,0)",xlab="Years",ylab="Rape Crime Rate Of Germany")
```

II. **Code for Kenya:**

```
K=read.csv("C:/Users/Dell/OneDrive/Desktop/new/Kenya.csv",header=TRUE)
K
colnames(K)
summary(K)
boxplot(K$Kenya,main="Figure-7
BOXPLOT FOR KENYA",ylab="Rape Crime Rate",col="yellow")
outliers3_rm<-K$Kenya[!K$Kenya%in% boxplot.stats(K$Kenya)$out]
length(K$Kenya)
length(outliers3_rm)
length(K$Kenya)-length(outliers3_rm)
boxplot(outliers3_rm,main="Figure-7.1
BOXPLOT FOR KENYA",ylab="Rape Crime Rate",col="yellow")
library(tseries)
library(forecast)
library(generics)
k_ts<-ts(outliers3_rm,start=c(2007),frequency=1)
k_ts
plot(k_ts,main="Figure-21 (Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of Kenya")
myarima=auto.arima(k_ts,trace=T,stepwise=F,approximation=F)
```

```
accuracy(myarima)
par(cex.axis=0.7)
autoplot(forecast(auto.arima(k_ts,trace=T,stepwise=F,approximation=F)),main="Figure-22
(Forecasting Graph)
Forecast from Arima(0,1,1)",xlab="Years",ylab="Rape Crime Rate Of Kenya")
forecast(auto.arima(k_ts,trace=T,stepwise=F,approximation=F))
```

III. Code for India:

```
IN=read.csv("C:/Users/Dell/OneDrive/Desktop/new/India.csv",header=TRUE)
IN
colnames(IN)
summary(IN)
boxplot(IN$India,main="Figure-4 BOXPLOT FOR INDIA",ylab="Rape Crime Rate",col="yellow")
library(tseries)
library(forecast)
library(generics)
in_ts<-ts(IN$India,start=2004,end=2022,frequency=1)
in_ts
plot(in_ts,main="Figure-23 (Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of India")
myarima=auto.arima(in_ts,trace=T,stepwise=F,approximation=F)
accuracy(myarima)
par(cex.axis=0.7)
autoplot(forecast(auto.arima(in_ts,trace=T,stepwise=F,approximation=F)),main="Figure-24
(Forecasting Graph)
Forecast from Arima(0,1,0)",xlab="Years",ylab="Rape Crime Rate Of India")
forecast(auto.arima(in_ts,trace=T,stepwise=F,approximation=F))
```

IV. Codes for the top 10 states:

a. Delhi:

```
D=read.csv("C:/Users/Dell/OneDrive/Desktop/new/finaledata.csv",header=TRUE)
D
attach(D)
summary(D)
 boxplot(D$Delhi,main="Figure-9 BOXPLOT FOR DELHI",col="purple",xlab="(i)",ylab="Rape Crime
Rate")
library(tseries)
library(forecast)
library(generics)
d_ts<-ts(D$Delhi,start=2001,end=2022,frequency=1)
d_ts
plot(d_ts,main="Figure-25 (Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of Delhi")
myarima=auto.arima(d_ts,trace=T,stepwise=F,approximation=F)
accuracy(myarima)
par(cex.axis=0.7)
autoplot(forecast(auto.arima(d_ts,trace=T,stepwise=F,approximation=F)),main="Figure-26
(Forecasting Graph)
Forecast from Arima(1,0,0) with zero mean",xlab="Years",ylab="Rape Crime Rate Of Delhi")
forecast(auto.arima(d_ts,trace=T,stepwise=F,approximation=F))
```

b. Mizoram:

```
D=read.csv("C:/Users/Dell/OneDrive/Desktop/new/finaledata.csv",header=TRUE)
D
```

```
attach(D)
colnames(D)
rownames(D)
summary(D)
boxplot(D$Mizoram,main="Figure-10
BOXPLOT FOR MIZORAM",col="purple",ylab="Rape Crime Rate")#outliers present
outliers_rm<-D$Mizoram[!D$Mizoram%in% boxplot.stats(D$Mizoram)$out]
length(D$Mizoram)
length(outliers_rm)
length(D$Mizoram)-length(outliers_rm)
boxplot(outliers_rm,main="Figure-10.1
BOXPLOT FOR MIZORAM",col="purple",ylab="Rape Crime Rate")
library(tseries)
library(forecast)
library(generics)
m_ts<-ts(outliers_rm,start=c(2007),end=c(2022),frequency=1)
m_ts
plot(m_ts,main="Figure-28(Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of Mizoram")
myarima=auto.arima(m_ts,trace=T,stepwise=F,approximation=F)
accuracy(myarima)
par(cex.axis=0.7)
autoplot(forecast(auto.arima(m_ts,trace=T,stepwise=F,approximation=F)),xlab="Years",ylab="Rap
e Crime Rate Of Mizoram")
forecast(auto.arima(m_ts,trace=T,stepwise=F,approximation=F))
```

c. **Madhya Pradesh:**

```
boxplot(D$Madhya.Pradesh,main="Figure-11 BOXPLOT FOR MADHYA
PRADESH",col="purple",ylab="Rape Crime Rate")
library(tseries)
library(forecast)
library(generics)
mp_ts=ts(D$Madhya.Pradesh,start=2001,end=2022,frequency=1)
mp_ts
plot(mp_ts,main="Figure-29 (Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of Madhya
Pradesh")
myarima=auto.arima(mp_ts,trace=T,stepwise=F,approximation=F)
accuracy(myarima)
par(cex.axis=0.7)
autoplot(forecast(auto.arima(mp_ts,trace=T,stepwise=F,approximation=F)),main="Figure-30
(Forecasting Graph) Forecast from Arima(0,0,2) with non-zero mean",xlab="Years",ylab="Rape
Crime Rate Of Madhya Pradesh")
forecast(auto.arima(mp_ts,trace=T,stepwise=F,approximation=F))
```

d. **Assam:**

```
boxplot(D$Assam,main="Figure-12
BOXPLOT FOR ASSAM",col="purple",ylab="Rape Crime Rate")
library(tseries)
library(forecast)
library(generics)
a_ts<-ts(D$Assam,start=2004,end=2022,frequency=1)
a_ts
```

```
plot(a_ts,main="Figure-31 (Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of Assam")
myarima=auto.arima(a_ts,trace=T,stepwise=F,approximation=F)
accuracy(myarima)
par(cex.axis=0.7)
autoplot(forecast(auto.arima(a_ts,trace=T,stepwise=F,approximation=F)),main="Figure-32
(Forecasting Graph) Forecast from Arima(0,1,0)",xlab="Years",ylab="Rape Crime Rate Of Assam")
forecast(auto.arima(a_ts,trace=T,stepwise=F,approximation=F))
```

e. **Arunachal Pradesh:**

```
boxplot(D$Arunachal.Pradesh,main="Figure-13 BOXPLOT FOR ARUNACHAL PRADESH",ylab="Rape
Crime Rate",col="purple")
library(tseries)
library(forecast)
library(generics)
ap_ts<-ts(D$Arunachal.Pradesh,start=2004,end=2022,frequency=1)
ap_ts
plot(ap_ts,main="Figure-33 (Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of Arunachal
Pradesh")
myarima=auto.arima(ap_ts,trace=T,stepwise=F,approximation=F)
accuracy(myarima)
par(cex.axis=0.7)
autoplot(forecast(auto.arima(ap_ts,trace=T,stepwise=F,approximation=F)),main="Figure-34
(Forecasting Graph) Forecast from Arima(0,1,0)",xlab="Years",ylab="Rape Crime Rate Of
Arunachal Pradesh")
forecast(auto.arima(ap_ts,trace=T,stepwise=F,approximation=F))
```

f. **Chhattisgarh:**

```
boxplot(D$Chhattisgarh,main="Figure-14 BOXPLOT FOR CHHATTISGARH",ylab="Rape Crime
Rate",col="purple")
library(tseries)
library(forecast)
library(generics)
c_ts<-ts(D$Chhattisgarh,start=2004,end=2022,frequency=1)
c_ts
plot(c_ts,main="Figure-35 (Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of
Chhattisgarh")
myarima=auto.arima(c_ts,trace=T,stepwise=F,approximation=F)
accuracy(myarima)
par(cex.axis=0.7)
autoplot(forecast(auto.arima(c_ts,trace=T,stepwise=F,approximation=F)),main="Figure-36
(Forecasting Graph) Forecast from Arima(0,1,0)",xlab="Years",ylab="Rape Crime Rate Of
Chhattisgarh")
forecast(auto.arima(c_ts,trace=T,stepwise=F,approximation=F))
```

g. **Rajasthan:**

```
boxplot(D$Rajasthan,main="Figure-15 BOXPLOT FOR RAJASTHAN",ylab="Rape Crime
Rate",col="purple")
library(tseries)
library(forecast)
library(generics)
r_ts<-ts(D$Rajasthan,start=2004,end=2022,frequency=1)
r_ts
```

```
plot(r_ts,main="Figure-37 (Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of Rajasthan")
myarima=auto.arima(r_ts,trace=T,stepwise=F,approximation=F)
accuracy(myarima)
par(cex.axis=0.7)
autoplot(forecast(auto.arima(r_ts,trace=T,stepwise=F,approximation=F)),main="Figure-38
(Forecasting Graph) Forecast from Arima(0,1,0) with drift",xlab="Years",ylab="Rape Crime Rate Of
Rajasthan")
forecast(auto.arima(r_ts,trace=T,stepwise=F,approximation=F))
```

h. **Tripura:**

```
boxplot(D$Tripura,main="Figure-16 BOXPLOT FOR TRIPURA",ylab="Rape Crime
Rate",col="purple")
library(tseries)
library(forecast)
library(generics)
t_ts<-ts(D$Tripura,start=2004,end=2022,frequency=1)
t_ts
plot(t_ts,main="Figure-39 (Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of Tripura")
myarima=auto.arima(t_ts,trace=T,stepwise=F,approximation=F)
accuracy(myarima)
par(cex.axis=0.7)
autoplot(forecast(auto.arima(t_ts,trace=T,stepwise=F,approximation=F)),main="Figure-40
(Forecasting Graph) Forecast from Arima(1,0,0) with non-zero mean ",xlab="Years",ylab="Rape
Crime Rate Of Tripura")
forecast(auto.arima(t_ts,trace=T,stepwise=F,approximation=F))
```

i. **Chandigarh:**

```
boxplot(D$Chandigarh,main="Figure-17 BOXPLOT FOR CHANDIGARH",ylab="Rape Crime
Rate",col="purple")
library(tseries)
library(forecast)
library(generics)
ch_ts<-ts(D$Chandigarh,start=2004,end=2022,frequency=1)
ch_ts
plot(ch_ts,main="Figure-41 (Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of
Chandigarh")
myarima=auto.arima(ch_ts,trace=T,stepwise=F,approximation=F)
accuracy(myarima)
par(cex.axis=0.7)
autoplot(forecast(auto.arima(ch_ts,trace=T,stepwise=F,approximation=F)),main="Figure-42
(Forecasting Graph) Forecast from Arima(0,1,1) with drift",xlab="Years",ylab="Rape Crime Rate Of
Chandigarh")
forecast(auto.arima(ch_ts,trace=T,stepwise=F,approximation=F))
```

j. **Haryana:**

```
boxplot(D$Haryana,main="Figure-18 BOXPLOT FOR HARYANA",ylab="Rape Crime
Rate",col="purple")
library(tseries)
library(forecast)
library(generics)
h_ts<-ts(D$Haryana,start=1,frequency=1)
h_ts
```

```
plot(h_ts,main="Figure-43 (Time Series Plot)",xlab="Years",ylab="Rape Crime Rate Of Haryana")
myarima=auto.arima(h_ts,trace=T,stepwise=F,approximation=F)
accuracy(myarima)
par(cex.axis=0.7)
autoplot(forecast(auto.arima(h_ts,trace=T,stepwise=F,approximation=F)),main="Figure-44
(Forecasting Graph) Forecast from Arima(0,1,0) with drift",xlab="Years",ylab="Rape Crime Rate Of
Haryana")
forecast(auto.arima(h_ts,trace=T,stepwise=F,approximation=F))
```

Reference

I. The links of the websites that are used in this project are as follows:

The data of rape crime rate against women in India for the years 2022:

- https://www.ncrb.gov.in/uploads/nationalcrimerecordsbureau/custom/1701935180TABLE3A2.pdf

The data of rape crime rate against women in India for the years 2021:

- https://www.ncrb.gov.in/uploads/nationalcrimerecordsbureau/post/1679652786TABLE3A2.pdf

The data of rape crime rate against women in India for the years 2020:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/TABLE-3A.2.pdf

The data of rape crime rate against women in India for the years 2019:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-3A.2_2.pdf

The data of rape crime rate against women in India for the years 2018:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-3A.2_0.pdf

The data of rape crime rate against women in India for the years 2017:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-3A.2_1.pdf

The data of rape crime rate against women in India for the years 2016:

- https://www.ncrb.gov.in/uploads/nationalcrimerecordsbureau/custom/1683013216Table3A2.pdf

The data of rape crime rate against women in India for the years 2015:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-5.2_2015.pdf

The data of rape crime rate against women in India for the years 2014:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-5.2_2014.pdf

The data of rape crime rate against women in India for the years 2013:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-5.2_2013.pdf

The data of rape crime rate against women in India for the years 2012:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table5.2.pdf

The data of rape crime rate against women in India for the years 2011:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-5.2_2011.pdf

The data of rape crime rate against women in India for the years 2010:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-5.2_2010.pdf

The data of rape crime rate against women in India for the years 2009:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-.2_2009.pdf

The data of rape crime rate against women in India for the years 2008:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-5.2_2008.pdf

The data of rape crime rate against women in India for the years 2007:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-5.2_2007.pdf

The data of rape crime rate against women in India for the years 2006:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-5.2_2006.pdf

The data of rape crime rate against women in India for the years 2005:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-5.2_2005.pdf

The data of rape crime rate against women in India for the years 2004:

- https://www.ncrb.gov.in/uploads/2022/July/11/custom/crime-in-india/Table-5.2-2004.pdf

The data of rape crime rate against women in Germany for the years 2004-2022:

- https://dataunodc.un.org/

The data of rape crime rate against women in Kenya for the years 2004-2022:

- https://dataunodc.un.org/

II. Fundamentals of Statistics (Vol II) - A.K. Gun, M.K Gupta, B Dasgupta.
III. An Outline of Statistical Analysis Theory (Voll II) - A.K. Gun, M.K Gupta, B Dasgupta.
IV. Introduction to Time Series Analysis and Forecasting- Douglas C. Montgomery, Chery L. Jennings, Murat Kulach.

Data Source

In context of analysing rape crime data for India, the data collection procedure involves meticulous steps to ensure accuracy, reliability and comprehensiveness. Below is an outline of procedure, specifically for collecting the rape crime data for India.

I. The data containing number of Rape Crime Rate cases for different factors (like age of the victims and relation with the offenders) for different States and Union Territories of India for the years 2000-2022 is obtained from the official website of National Crime Records Bureau (NCRB), Ministry of Home Affairs, India.
II. The data containing Rape Crime Rate reported under total rape section 376 IPC for different States and Union Territories of India for the years 1997-2022 is obtained from the official website of National Crime Records Bureau (NCRB), Ministry of Home Affairs, India.
III. The data containing total reported Rape Crime Rate cases of Germany from 2003 to 2022 years which is obtained from the website of UNDOC (United Nation Office of Drugs and Crime).
IV. The data containing total reported Rape Crime Rate cases of Kenya from 2004 to 2022 years which is obtained from the website of UNDOC (United Nation Office of Drugs and Crime).

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Abstract
This study undertakes a comparative, on statistical ground for one of the most pressing and sensitive social issues faced world-wide that is rape. Focusing on three diverse countries, namely India, Germany and Kenya, the study aims to analyse patterns, trends, and societal impacts by utilizing official crime statistics data drawn from national and international crime reports. Using time series analysis and comparative data visualization techniques, the project evaluates the reported rape cases across these countries over the past two decades, measuring rates per 100,000 population and examining year-on-year growth or decline.
Within India, the study delves deeper by exploring rape crime rates across various states and union territories. Based on an average analysis of the past 19 year's data, we choose 10 states (i.e.; Delhi, Mizoram, Madhya Pradesh, Assam, Arunachal Pradesh, Chhattisgarh, Rajasthan, Tripura, Chandigarh and Haryana) that consistently reflect high case volumes due to rape crime. It further provides predictive insights into the future trends of reported rape cases for the next 10 years (2023 to 2032) for these Indian states and also for Germany, Kenya and India based on their total rape crime rate.
Beyond mere statistical analysis, this research also seeks to shed light on the underlying systemic and structural issues contributing to the persistent prevalence of rape. By combining statistical forecasting with socio-cultural critique, the study emphasizes the urgent need to address gender-based violence on a global scale, providing critical insights into how this crime is evolving over time and the societal transformations and strict laws required to combat it.

Introduction
General Legal definition of Rape: Rape is the act of engaging in sexual intercourse or other forms of sexual penetration with another person without their consent, typically through force, threats, manipulation or when the person is incapable of giving consent due to age, mental incapacity, or intoxication.
Rape is one of the most heinous and traumatic crimes that deeply scars not only the victim but also the collective conscience of society. This research is to draw attention towards the globally increased Rape Crime in the world. By a comparative study among three countries of the world we aim to find if this crime is regional, economical, educational or societal based. Firstly, Germany (a European country) which is considered to be a developed country in economic field. Secondly, Kenya (an African country) which is considered to be a lower- or middle-income country in economic field. Thirdly, India (an Asian country) which is considered to be a developing or lower- or middle-income country in economic field and most importantly it's our nationality. While there has been growing discourse around women's safety in India, there remains a gap in long-term predictive research that can aid proactive policy and societal response. Moreover, considering mainly India, this project further explores with the Top 10 States and Union Territories dealing with their Rape Crime Rates, comparing the states with each other. Divining into the brief details of India, we see despite the existence of strict laws and increasing awareness around women's rights and safety, rape remains a grave and persistent issue. The country has and continues witness a troubling rise in reported rape cases over the years, highlighting deep-rooted issues such as gender inequality, social stigma,

lack of education and gaps in law enforcement. The issue gained global attention after the 2012 Nirbhaya gang rape case in Delhi, which led to widespread protests and demands for reform. In response, the Indian government introduced several legal changes, including stricter punishments and the establishment of fast-track courts. However, the prevalence of such crimes continues and many cases still go unreported due to fear of social ostracism, victim-blaming, and lack of faith in the justice system.

Objective

The primary objective of this research is to address one of the most alarming issues faced world-wide by women today that is the prevalence of rape. Analyse and compare the rape crime data of 3 countries (India, Germany & Kenya) across the world. It mainly compares the various (Top 10) states in India using forecasting models such as time series plots, data visuals and ARIMA (Auto-Regressive Integrated Moving Average) by doing so, the study seeks to:

I. To Analyse Historical Trends of Rape Cases: The study examines the year-wise data on reported rape cases of the TOP 10 Indian states and union territories. Also, examines the global trend year-wise of total cases for 3 countries. Identifies patterns, fluctuations, and significant spikes or drops in reported cases for these countries.

II. Identify State-wise Hotspots of Gender-based Violence: Helps to highlight the regions with consistently high or increasing numbers of rape cases and compares the state performances based on normalized crime rates (cases per lakh population) which helps to understand socio-demographic factors contributing to regional disparities.

III. To Forecast Rape Cases for the Next 10 Years (2023–2032): We have used time series models like ARIMA to forecast the number of rape cases in each of the (TOP 10) states. The prediction of long-term trends helps to assess whether the situation is likely to improve or worsen without significant intervention.

IV. Data Visualisation causes public awareness: Through interactive visualizations and accessible analysis, the project can educate the public about the ongoing and projected state of women's safety in India. It can help shift the conversation from reactionary outrage to sustained, preventive action.

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